

Stakeholder Question Catalogue 02

Taught Postgraduate Students

80 taught-postgraduate questions mapped to feasible sensing, authoritative institutional data, analysis methods, uncertainty and safe answer boundaries.

STAKEHOLDER	Students on taught postgraduate and professional programmes using Abacws for intensive modules, seminars, independent and dissertation study, specialist computing or project work, group assessment, presentations and mixed campus schedules.
QUESTION RANGE	Immediate status, historical comparison, prediction, diagnosis, recommendation, itinerary optimisation, accessibility, service continuity, sustainability, permissions, evidence transparency and safety-bounded interaction.
EVIDENCE MODEL	Proposed room-level sensing combined with authorised timetable, booking, access, spatial, BMS, energy, network, AV, maintenance, accessibility, weather and institutional service data.

Core principle: support the student decision while stating what is observed, looked up, calculated, estimated, forecast, diagnosed, recommended, restricted or not assessable.

1. Catalogue scope and design principles

The catalogue focuses on decisions that taught postgraduate students may reasonably want a conversational university building to support during intensive teaching, independent study, specialist work and assessment peaks.

Proposed capability, not current fact

Every sensor, integration and service is a design requirement unless deployment, coverage, permissions and operational status are verified. The catalogue does not claim that all capabilities currently exist.

Current building information

The May 2021 floor plan is a historical spatial reference. Current room use, capacity, student access, bookability, equipment, entrances, lifts, services and HVAC relationships require authoritative verification.

Postgraduate decision context

Questions reflect intensive modules, seminars, dissertation and deep work, specialist computing or project activity, group assessment, deadline peaks, commuting, part-time and international study patterns.

Question-to-evidence design

Each answer resolves the decision, place, period and hard constraints; identifies minimum evidence; checks source quality and authority; performs the correct operation; and states uncertainty.

Authoritative source precedence

Bookings, access, timetable, room purpose, lifts, closures, network, AV, maintenance, digital services and emergencies come from authorised operational systems, not environmental inference.

Data-quality and coverage gating

Location, calibration or commissioning, timestamp, uptime, missingness, drift, comparability and source disagreement are checked before any room claim, forecast, diagnosis or recommendation.

Privacy, accessibility and permissions

Use anonymous aggregate occupancy and numerical sound statistics, never speech or identity. Apply student permissions and treat accessibility, safety and authorised-use constraints as hard filters.

Safe uncertainty and refusal

Forecasts are probabilistic, diagnoses rank evidence rather than invent causes, recommendations need fallbacks, and missing or conflicting critical evidence should produce clarification, a limited answer or not assessable.

2. How to read the catalogue

Readiness describes the verified sensing, institutional integrations, validation and governance needed for a responsible answer; it does not rank educational value.

R1 - practical with core feeds

Plausible once verified spatial, access, service or sensor metadata and the required core authoritative feeds are available and quality-controlled.

R2 - integration-dependent

Requires additional timetable, booking, BMS, network, AV, digital-service, lift, maintenance, weather or forecasting integration.

R3 - research-grade or restricted

Requires dense or specialist sensing, validated diagnosis or prediction, operationally restricted data, or stronger privacy, security and safety governance.

Answer depth and operation types

- L1** Authoritative lookup or current observation from a verified source.
- L2** Calculation, aggregation or bounded historical comparison.
- L3** Estimate or forecast combined with a recommendation and uncertainty.
- L4** Diagnosis, optimisation, intervention evaluation or evidence audit.

Question sections

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GLOBAL ANSWER RULES

Use only verified student-accessible candidates; preserve source timestamps, permissions and room-level coverage; use anonymous occupancy and numerical sound statistics; distinguish observation, lookup, calculation, estimate, forecast, diagnosis and recommendation; defer bookings, access, digital services, alarms and closures to authoritative systems; and return a limited or not-assessable answer when critical evidence is missing.

3. Worked example: mixed postgraduate study day

A responsible answer must combine the student schedule and hard constraints with verified access, room purpose, service status, room-level evidence and uncertainty; it must not treat an empty or quiet space as automatically available or private.

Student: "I am on campus next Thursday from 10 a.m. to 7 p.m. I have a seminar, a group-presentation rehearsal and an online supervision meeting. Where should I work between them so I can write my dissertation, join the call appropriately and avoid unnecessary travel?"

1. Resolve the study day

Resolve times, durations, locations, dissertation-work needs, group size, call privacy, equipment, step-free access and the latest acceptable arrival at each activity.

4. Forecast each interval

Use comparable teaching days, scheduled activity and current trends to forecast crowding, sound and room conditions; state which services are live lookups rather than predictions.

2. Verify authorised candidates

Use the current space registry, permitted use, access, bookings, opening hours and closures; do not infer availability, privacy or specialist capability from occupancy or the 2021 plan.

5. Optimise with contingencies

Apply hard access, privacy and equipment constraints first; minimise travel; rank a primary and backup space for each interval; avoid backups dependent on the same unverified service.

3. Verify evidence and services

Require room-level occupancy and sound coverage, environmental freshness, aggregate network or AV status and evidence-quality metadata; preserve the authority of booking and service systems.

6. Communicate answerability

State the recommendation, evidence period, confidence, omitted criteria, closing times, refresh point and switch trigger; ask for clarification or return not assessable when a critical source is missing.

ILLUSTRATIVE ANSWER STRUCTURE

"For your first study interval, [verified space] is the strongest option because it meets confirmed access, activity and service requirements and is forecast to have lower crowding and sound. For the supervision call, use [verified enclosed or bookable room]; quietness alone is not evidence of privacy. [Backup] is available if conditions change. Confidence is [low, moderate or high] because [coverage and schedule uncertainty]. Recheck at [time]. [Criterion] was not used because [missing, stale or restricted source]."

1. Intensive module and weekly planning

Questions PGT-001-PGT-002

PGT-001

CORE

L3 - prediction and recommendation

R2 - integration-dependent

Taught postgraduate question: I have a 3 p.m. seminar in Abacws on Thursday and want two hours beforehand for dissertation writing. Where am I most likely to find a quiet, lightly used student space with power and working Wi-Fi, while still leaving enough time to reach the seminar?

Sensors and telemetry

Design requirement: room-level temperature, RH, direct NDIR CO₂, task-area illuminance and locally computed LAeq/L10/L90; anonymous room occupancy and corridor footfall; aggregate network-service and room electrical-fault status. Require verified room/route mapping, synchronised timestamps, health and gap flags; corridor evidence is contextual and network telemetry remains aggregate.

Authoritative supporting data

Authoritative current sources: student-accessible space inventory, present room use and permitted activity, booking status, capacity, power provision, opening hours, route graph, seminar room and time, network incident/status feed and only the schedule details the student submits. Verify all current room facts operationally; the May 2021 plan is spatial context only.

Analysis required

Forecast + recommendation: resolve the two-hour study window and latest safe departure; forecast occupancy, sound and measured room conditions from comparable Thursdays and scheduled activity; filter by access, power and network status; rank a primary and backup option with confidence and a recheck time.

Answer boundary

This is a forecast, not a guaranteed seat, socket or device-specific network result. Use only the schedule details supplied for the request. Exclude candidates with unverified access, booking, service or room-level evidence. If comparable history or current coverage is insufficient, give a broad range or decline to recommend.

Why this is useful: Makes a short pre-seminar study window more dependable and gives the student a realistic fallback before a fixed class time.

PGT-002

CORE

L4 - itinerary optimisation

R2 - integration-dependent

Taught postgraduate question: Tomorrow I have classes on different floors, a short online meeting and two gaps for reading. Can you plan a practical sequence of spaces that keeps unnecessary movement to a minimum?

Sensors and telemetry

Design requirement: anonymous room occupancy, corridor/stair footfall, locally computed sound statistics, room temperature, CO₂ and illuminance, plus official lift and aggregate network status. No personal tracking or raw audio. Require verified room/route mapping, synchronised timestamps, current health and gap flags; unobserved segments remain unknown.

Authoritative supporting data

Authoritative current sources: timetable supplied with consent, verified space uses and student permissions, bookings, room features, opening hours, route graph, lift and closure status, and online-meeting requirements stated by the student. Verify current room use, access, bookability and route status in authorised operational records; the May 2021 plan is spatial context only.

Analysis required

Forecast + itinerary recommendation: divide the day into class, reading and online-meeting intervals; treat access, step-free needs, call suitability and transfer time as hard constraints; forecast crowding and room conditions; minimise unnecessary movement and provide a time-feasible backup for each gap.

Answer boundary

The system must not infer attendance, monitor movement between sessions or reserve rooms. A route segment with unknown lift, door or closure status cannot be labelled accessible. Recommendations must be refreshed before each interval; if a critical feed is unavailable, provide only the verified part of the itinerary.

Why this is useful: Helps a taught postgraduate manage a compressed day of classes, study and online activity in a six-level building.

1. Intensive module and weekly planning

Questions PGT-003-PGT-004

PGT-003

CORE

L3 - multi-activity recommendation

R2 - integration-dependent

Taught postgraduate question: Which day this week gives me the best chance of finding four hours for focused study followed by a one-hour project-group meeting?

Sensors and telemetry

Historical and near-term room occupancy, corridor footfall, LAeq/L10/L90 sound statistics, temperature, CO2 and illuminance for monitored candidate areas. Quality gate: use only commissioned or calibrated points with verified room or served-zone mapping, synchronised timestamps, current health and explicit missing-data flags. Corridor or adjacent-room readings are context only, never a room substitute.

Authoritative supporting data

Authoritative current sources: student-accessible spaces, bookings, timetable and event schedule, opening hours, room capacities and permitted activities; group availability only when members choose to provide it; weather forecast. Verify current room use, access, bookability and route status in authorised operational records; the May 2021 plan is spatial context only.

Analysis required

Forecast plus recommendation: identify valid days and consecutive study-plus-meeting windows; forecast quiet-study and group-space demand separately; assess environmental stability and transition time; return several options with uncertainty and the number of comparable weeks.

Answer boundary

Do not infer group members' schedules or guarantee room availability. Unmonitored or rarely used spaces must be labelled unknown rather than treated as inferior, and group consent is required for shared schedule data. When history is sparse, sensing coverage has changed or future schedules are not comparable, widen the prediction interval or withhold the forecast.

Why this is useful: Supports efficient planning when a taught postgraduate has limited time on campus and needs two different kinds of space.

PGT-004

CORE

L3 - peak-demand forecast

R2 - integration-dependent

Taught postgraduate question: During next week's assessment peak, when should I arrive for the best chance of finding a suitable three-hour study space?

Sensors and telemetry

Historical anonymous aggregate occupancy, entrance and doorway counts, corridor footfall, study-area sound statistics and room environmental measurements, with sensor uptime and coverage records. Quality gate: use only commissioned or calibrated points with verified room or served-zone mapping, synchronised timestamps, current health and explicit missing-data flags. Corridor or adjacent-room readings are context only, never a room substitute.

Authoritative supporting data

Authoritative current sources: assessment calendar, timetable, bookings, opening hours, room capacities, current access rules and planned events. Verify current room use, access, bookability, equipment and services in authorised operational records; the May 2021 plan supplies spatial context only.

Analysis required

Forecast: identify comparable assessment and deadline periods, model candidate-space demand by arrival time, exclude closed or restricted spaces and return the probability of finding a broadly suitable space with a later fallback window.

Answer boundary

Anonymous occupancy estimates do not prove that a desk, socket or workstation is free. Changed access, room stock or sensor coverage must be modelled explicitly. Do not identify students or infer academic performance; when comparable history is inadequate, report that a reliable forecast is not assessable.

Why this is useful: Helps students plan arrival during short, high-pressure assessment periods rather than searching after all suitable areas are busy.

1. Intensive module and weekly planning

Questions PGT-005-PGT-006

PGT-005

SITUATIONAL

L2 - historical comparison

R2 - integration-dependent

Taught postgraduate question: When I come to campus, are mornings or evenings usually better for low noise, reliable access, working network services and stable room conditions?

Sensors and telemetry

Historical room-level LAeq/L10/L90 sound statistics, anonymous aggregate occupancy, temperature, CO2, illuminance and authorised aggregate network-service health; entrance, lift and door status history where authorised. Quality gate: use only commissioned or calibrated points with verified room or served-zone mapping, synchronised timestamps, current health and explicit missing-data flags. Corridor or adjacent-room readings are context only, never a room substitute.

Authoritative supporting data

Authoritative current sources: opening and access rules, booked events, timetable, planned closures, security-service status and current student permissions. Verify current room use, access, bookability, equipment and services in authorised operational records; the May 2021 plan supplies spatial context only.

Analysis required

Historical comparison + recommendation: compare equivalent morning and evening windows over several weeks using median and upper-percentile crowding, sound, service incidents and environmental variability; let the student weight travel, access and study preferences.

Answer boundary

The result concerns monitored building conditions, not personal safety or a guarantee of network availability. Evening options require a same-day check of opening, access, lift and route status. If the periods are not comparable, state the limitation and avoid a single-cause explanation.

Why this is useful: Helps part-time and commuting students deciding when a limited campus visit is most productive.

PGT-006

CORE

L3 - recurring schedule recommendation

R2 - integration-dependent

Taught postgraduate question: Across the remaining teaching weeks, which recurring two-hour period is most dependable for my weekly module reading?

Sensors and telemetry

Historical, correctly located room-level LAeq/L10/L90, anonymous aggregate occupancy, corridor footfall, temperature, CO2 and illuminance, with missingness and coverage metadata. Quality gate: use only commissioned or calibrated points with verified room or served-zone mapping, synchronised timestamps, current health and explicit missing-data flags. Corridor or adjacent-room readings are context only, never a room substitute.

Authoritative supporting data

Authoritative current sources: timetable, room bookings, events, opening hours and verified study-space access for the remaining teaching weeks. Verify current room use, access, bookability, equipment and services in authorised operational records; the May 2021 plan supplies spatial context only.

Analysis required

Forecast plus recommendation: identify recurring candidate windows, compare their median and worst-case conditions, check whether future timetable patterns differ from history, and rank robust options rather than the single quietest past period.

Answer boundary

Sound statistics cannot reveal conversation content or guarantee silence. If future scheduled use differs materially from the historical period, confidence must be reduced and the system should avoid presenting a fixed routine as certain. When history is sparse, sensing coverage has changed or future schedules are not comparable, widen the prediction interval or withhold the forecast.

Why this is useful: Helps a taught postgraduate establish a sustainable routine within a short academic term.

1. Intensive module and weekly planning

Questions PGT-007-PGT-008

PGT-007

CORE

L2 - short-gap recommendation

R1 - practical with core feeds

Taught postgraduate question: I have only 45 minutes between sessions. Which nearby space would give me useful study time without putting the next class at risk?

Sensors and telemetry

Live anonymous aggregate occupancy, corridor footfall, room-level LAeq/L10/L90 sound statistics and current environmental measurements for monitored candidate spaces. Verify the current route graph and the timestamp, health and coverage of each lift, door, closure and aggregate-footfall source used; do not treat an unobserved route segment as uncongested or accessible.

Authoritative supporting data

Authoritative current sources: the next room supplied by the student, verified student-accessible spaces, current bookings, route graph, lift status and temporary closures. Verify current room use, access, bookability, equipment and services in authorised operational records; the May 2021 plan supplies spatial context only.

Analysis required

Current-state lookup plus recommendation: calculate usable study time after walking and setup; filter spaces by access and current use; rank nearby options by live crowding, sound and route reliability; include a departure time and backup.

Answer boundary

Do not track the student or guarantee a place. Route and lift information must be current, and the system should recommend leaving early when congestion or data uncertainty makes the transfer risky. Exclude every candidate with an unverified hard constraint; if no valid option remains, say so and provide the authorised next step rather than guess.

Why this is useful: Makes a short timetable gap useful without compromising punctuality.

PGT-008

SITUATIONAL

L3 - delay-aware recommendation

R2 - integration-dependent

Taught postgraduate question: My train is delayed. Given my new arrival time, where could I work briefly before class without being late?

Sensors and telemetry

Live room occupancy, corridor footfall, sound statistics and measured room conditions; current lift and entrance telemetry only through authorised systems. Verify the current route graph and the timestamp, health and coverage of each lift, door, closure and aggregate-footfall source used; do not treat an unobserved route segment as uncongested or accessible.

Authoritative supporting data

Authoritative current sources: revised arrival time provided by the student, class location and start time, live bookings, current access and closure status, route graph and optional external travel status. Verify current room use, access, bookability, equipment and services in authorised operational records; the May 2021 plan supplies spatial context only.

Analysis required

Forecast and recommendation: short-horizon estimate plus recommendation: recalculate the usable window, exclude spaces that create lateness risk, compare live and forecast conditions, and return a primary and fallback location with a latest-departure time.

Answer boundary

The building cannot determine academic late-arrival policy or notify a lecturer unless an authorised service explicitly supports it. External travel data and room availability may change after the answer. When history is sparse, sensing coverage has changed or future schedules are not comparable, widen the prediction interval or withhold the forecast.

Why this is useful: Helps a commuter recover useful study time when a journey disruption compresses the day.

1. Intensive module and weekly planning

Questions PGT-009-PGT-010

PGT-009

SITUATIONAL

L3 - mixed-use planning

R2 - integration-dependent

Taught postgraduate question: I need a quiet lunch break and then 30 minutes to review slides before an evening seminar. Which sequence of spaces would work best?

Sensors and telemetry

Anonymous aggregate occupancy, corridor footfall, room-level LAeq/L10/L90 sound statistics, temperature, CO2 and illuminance in verified waiting, break and study spaces. Verify the current route graph and the timestamp, health and coverage of each lift, door, closure and aggregate-footfall source used; do not treat an unobserved route segment as uncongested or accessible.

Authoritative supporting data

Authoritative current sources: public and student-accessible space registry, permitted use, bookings, opening hours, seminar location, route graph, amenity information and temporary closures. Verify current room use, access, bookability, equipment and services in authorised operational records; the May 2021 plan supplies spatial context only.

Analysis required

Forecast and recommendation: sequence recommendation: identify appropriate break and study candidates, forecast their conditions for consecutive windows, minimise transfer time and avoid spaces whose permitted use conflicts with eating or quiet study.

Answer boundary

Do not infer dietary needs or claim that an area permits food without an authoritative rule. Recommendations must not expose staff-only spaces or rely on unverified amenity information. When history is sparse, sensing coverage has changed or future schedules are not comparable, widen the prediction interval or withhold the forecast.

Why this is useful: Supports a realistic evening-study routine that includes recovery time as well as academic preparation.

PGT-010

ADVANCED

L4 - demand-pattern analysis

R2 - integration-dependent

Taught postgraduate question: During this teaching block, when is Abacws usually busiest, and which scheduled activities seem to explain the peaks?

Sensors and telemetry

Historical entrance counts, corridor footfall and anonymous aggregate room occupancy, with coverage and sensor-change records. Quality gate: use only commissioned or calibrated points with verified room or served-zone mapping, synchronised timestamps, current health and explicit missing-data flags. Corridor or adjacent-room readings are context only, never a room substitute.

Authoritative supporting data

Authoritative current sources: teaching-block dates, timetable, bookings, events, room capacities, opening hours and planned closures. Verify current room use, access, bookability, equipment and services in authorised operational records; the May 2021 plan supplies spatial context only.

Analysis required

Historical pattern analysis + forecast: estimate aggregate demand by day and time, identify recurring peaks, test their association with authorised schedule and event categories, and forecast forthcoming peaks with intervals that reflect calendar and coverage changes.

Answer boundary

Keep results aggregate. Do not infer attendance by named student, cohort or module unless the schedule relationship is authorised and necessary. A temporal association with scheduled activity is not proof that one group caused a peak.

Why this is useful: Helps taught postgraduate students plan limited campus time around predictable demand peaks.

2. Seminars, workshops and assessed sessions

Questions PGT-011-PGT-012

PGT-011

CORE

L3 - class-condition forecast

R3 - research-grade or restricted

Taught postgraduate question: Our seminar lasts three hours and about 30 people are expected. Are the room's temperature and CO2 likely to remain reasonably stable throughout?

Sensors and telemetry

Design requirement: at least two correctly sited in-room NDIR CO2 and temperature/RH points for a 30-person seminar, anonymous doorway count or aggregate occupancy, serving-zone supply airflow and fan/damper state, and outdoor temperature. Require verified room/HVAC mapping, calibration or commissioning, synchronised clocks, current health and complete session coverage.

Authoritative supporting data

Authoritative current sources: confirmed room, scheduled duration and expected attendance, room capacity and present use, HVAC-to-room relationships, current bookings, relevant setpoints, outdoor weather and any planned plant override. Verify current room use, access, bookability and route status in authorised operational records; the May 2021 plan is spatial context only.

Analysis required

Forecast: establish pre-class conditions, room volume, expected occupancy and verified ventilation state; use a validated mass-balance or back-tested time-series model to predict zone-level CO2 and temperature trajectories with intervals; identify review windows rather than a single precise threshold time.

Answer boundary

This predicts environmental indicators, not individual comfort, health or regulatory compliance. Distributed in-room coverage is required; one node or a corridor sensor cannot characterise the seminar. If room volume, attendance, airflow or model validation is inadequate, provide current observations and a limited trend only.

Why this is useful: Helps the lecturer and students anticipate when a long, intensive seminar may need ventilation review or a short break.

PGT-012

SITUATIONAL

L2 - seating-zone comparison

R3 - research-grade or restricted

Taught postgraduate question: If the seminar is full, is there a broad part of the room that usually has lower background noise and a slower rise in CO2?

Sensors and telemetry

Required (proposed unless verified): at least two or more spatially distributed in-room NDIR CO2, temperature and numerical sound-measurement points appropriate to room geometry; anonymous occupancy distribution or doorway counts; supply-air terminal or zone airflow where available. Verify placement, calibration/commissioning, clock alignment and coverage; do not retain speech or identify seats or people.

Authoritative supporting data

Authoritative current sources: verified room geometry and seating zones, current seminar use and capacity, supply/return locations or serving-HVAC zone, expected attendance, door locations and any accessibility seating requirements supplied by the student. The May 2021 drawing is historical geometry only, not proof of current seating or services.

Analysis required

Observed and historical zone comparison: compare front/middle/rear or other broad validated zones during comparable full sessions; calculate distributions and uncertainty for sound and CO2 rise; recommend a broad area only when the spatial differences are repeatable.

Answer boundary

Do not recommend a precise seat, claim speech privacy or treat a few centimetres of difference as meaningful. If distributed sensing or comparable sessions are absent, report that zone-level advice is not assessable.

Why this is useful: Helps a student choose a broad area that better supports listening and sustained attention.

2. Seminars, workshops and assessed sessions

Questions PGT-013-PGT-014

PGT-013

CORE

L2 - pre-session readiness check

R2 - integration-dependent

Taught postgraduate question: I present in 40 minutes. Is the booked room ready to use, and are the display, network and room power reporting normally? What backup should I use if something is unavailable?

Sensors and telemetry

Required operational telemetry: authorised room-controller or AV-management self-test for display, input path and any installed conferencing endpoint; network-service and access-point health; room/circuit power status where instrumented; door/access state and current room-condition observations only as secondary context. Use time-stamped controller or incident feeds; do not infer readiness from occupancy.

Authoritative supporting data

Authoritative current sources: the confirmed booking, permitted presenter access, current equipment inventory, supported connection methods, IT/AV incident status, support contact, room-opening state, accessibility requirements and an approved backup room or presentation method. Current room use and equipment must be verified operationally, not from the May 2021 plan.

Analysis required

Authoritative lookup + readiness assessment: check the booking, access and each required room function against current inventory, self-test/controller status and incident feeds; classify each as ready, degraded, unavailable or unverified; return an approved backup and support route.

Answer boundary

Room telemetry cannot guarantee the student's laptop, account, adapter, external platform or presentation file. Do not inspect content or credentials. If an essential function has no current authoritative status, report it as unverified rather than declaring the room ready.

Why this is useful: Reduces avoidable presentation failure and gives the student time to move or contact support.

PGT-014

CORE

L2 - matched historical comparison

R2 - integration-dependent

Taught postgraduate question: Were last week's workshop conditions unusually warm, poorly ventilated or noisy compared with other sessions of the same module?

Sensors and telemetry

Room temperature, relative humidity, direct CO2, privacy-preserving, locally computed sound-level statistics, anonymous aggregate occupancy and HVAC airflow for the session and comparison periods; outdoor weather. Quality gate: use only commissioned or calibrated points with verified room or served-zone mapping, synchronised timestamps, current health and explicit missing-data flags. Corridor or adjacent-room readings are context only, never a room substitute.

Authoritative supporting data

Authoritative current sources: session times, room, expected attendance, module timetable and any recorded room or system changes. Verify current room use, access, bookability, equipment and services in authorised operational records; the May 2021 plan supplies spatial context only.

Analysis required

Matched historical comparison: select comparable sessions by room, duration, attendance and outdoor conditions; calculate level, duration and spatial variation; disclose missing data and whether the difference exceeds normal variability.

Answer boundary

The analysis describes measured conditions, not how every participant felt. Do not compare sessions with materially different sensing, attendance or operation, and do not attribute a difference to a person or teaching style. If comparability is weak, report the observations only.

Why this is useful: Provides evidence for a well-founded discussion with the module team or facilities staff.

2. Seminars, workshops and assessed sessions

Questions PGT-015-PGT-016

PGT-015

ADVANCED

L4 - multi-source diagnosis

R3 - research-grade or restricted

Taught postgraduate question: The workshop became much warmer after the break, when everyone returned and opened laptops. What evidence best explains the change?

Sensors and telemetry

Required telemetry (proposed unless verified): distributed in-room air temperature/relative humidity and, where justified, globe temperature; NDIR CO2; anonymous aggregate occupancy; room or relevant circuit power for computing load; serving-zone supply airflow, fan, valve and damper command/feedback; outdoor temperature and solar radiation. Use time-synchronised, quality-flagged streams with verified room/HVAC/meter topology and explicit gaps.

Analysis required

Diagnosis: reconstruct the break and restart at a common time base; quantify temperature-rate change, occupancy, plug-load and HVAC response; compare with matched workshops; test plausible heat-balance explanations; rank supported hypotheses and show which variables were not measured.

Authoritative supporting data

Authoritative current sources: workshop timeline, room and HVAC topology, equipment use policy, current setpoints, maintenance events and sensor placement metadata. Verify current room use, access, bookability, equipment and services in authorised operational records; the May 2021 plan supplies spatial context only.

Answer boundary

The result is a ranked diagnosis, not proof that laptops, people or HVAC alone caused the change. Do not estimate individual heat contribution or a mechanical fault without the relevant telemetry. If load, airflow, radiant or occupancy evidence is missing, report the observed sequence and remaining hypotheses only.

Why this is useful: Helps distinguish a normal high-load effect from a control, ventilation or equipment problem.

PGT-016

CORE

L3 - alternative-room recommendation

R2 - integration-dependent

Taught postgraduate question: Our seminar has to move. Which available room and time best fit the group size, step-free access, presentation equipment and likely room conditions?

Sensors and telemetry

Forecast room occupancy, temperature, CO2, illuminance and numerical sound conditions for monitored candidates; lift and door status through authoritative systems. Quality gate: use only commissioned or calibrated points with verified room or served-zone mapping, synchronised timestamps, current health and explicit missing-data flags. Corridor or adjacent-room readings are context only, never a room substitute.

Analysis required

Multi-criteria recommendation: apply hard requirements first, forecast conditions for candidate room-time pairs, account for travel between adjacent sessions, and explain the trade-offs and confidence of the top options.

Authoritative supporting data

Authoritative current sources: group size, timetable constraints, bookings, current room capacities and permitted use, accessibility, AV inventory, opening hours, route graph and planned events. Verify current room use, access, bookability, equipment and services in authorised operational records; the May 2021 plan supplies spatial context only.

Answer boundary

Do not assume that a room shown on the May 2021 plan is currently a seminar room, accessible or bookable. The result is a recommendation and cannot create or modify a booking without permission. When history is sparse, sensing coverage has changed or future schedules are not comparable, widen the prediction interval or withhold the forecast.

Why this is useful: Supports a defensible rescheduling choice rather than selecting the first nominally free room.

2. Seminars, workshops and assessed sessions

Questions PGT-017-PGT-018

PGT-017

SITUATIONAL

L3 - break-timing recommendation

R3 - research-grade or restricted

Taught postgraduate question: During our three-hour class, when are CO2 or temperature most likely to worsen enough for the lecturer to consider a short break?

Sensors and telemetry

Required telemetry (proposed unless verified): distributed in-room NDIR CO2 and air-temperature/relative humidity points, anonymous aggregate occupancy or doorway counts, and serving-zone airflow/fan status. Use verified room and HVAC mappings, synchronised clocks, current health, spatial coverage and missing-data flags; a corridor sensor is not a class-room measurement.

Authoritative supporting data

Authoritative current sources: class duration, scheduled attendance, room capacity and use, timetable, lecturer or organiser contact route, and any approved teaching-break policy. Verify current room use, access, bookability and route status in authorised operational records; the May 2021 plan is spatial context only.

Analysis required

Forecast and advisory calculation: model the indicator trajectory from pre-class conditions, expected attendance, room volume and ventilation state; back-test on comparable classes; report a prediction interval and a review window rather than a precise break minute; route the evidence to the authorised organiser.

Answer boundary

CO2 and temperature are operational indicators, not a complete measure of air quality, comfort or health. The lecturer or authorised organiser decides whether to pause teaching. If distributed room sensing, attendance or airflow evidence is inadequate, provide live observations only.

Why this is useful: Offers a practical way to manage long intensive sessions while exposing the evidence behind the suggestion.

PGT-018

SITUATIONAL

L2 - assessed-session condition check

R2 - integration-dependent

Taught postgraduate question: I have an assessed in-class test. Are there any current room or service issues I should know about, and where can I check approved adjustment details?

Sensors and telemetry

Design requirement: current in-room temperature, RH, direct NDIR CO2, illuminance and numerical sound statistics where deployed; official lift, power, network and AV status only when relevant to the assessment. Require verified room mapping, synchronised timestamps, health and gap flags; network evidence remains aggregate.

Authoritative supporting data

Authoritative current sources: official assessment room and time, access and closure status, invigilation or module notices, approved adjustment service, room features and support route. Personal adjustment details remain in the authorised student system. Verify current room use, access, bookability and route status in authorised operational records; the May 2021 plan is spatial context only.

Analysis required

Authoritative lookup + bounded observation: confirm the official assessment room/time, access and reported service incidents; summarise correctly located current room measurements with timestamps and quality flags; keep general room evidence separate from the student's approved adjustments.

Answer boundary

Do not expose or infer disability, health information, adjustment details or likely performance. Official assessment instructions and the authorised adjustments service take precedence. Missing or stale evidence must be labelled 'unknown', not converted into a suitability claim.

Why this is useful: Reduces uncertainty before an assessed session and directs the student to the correct official support route when needed.

2. Seminars, workshops and assessed sessions

Questions PGT-019-PGT-020

PGT-019

SITUATIONAL

L4 - class-change congestion analysis

R2 - integration-dependent

Taught postgraduate question: Do queues near the room regularly delay this seminar, and how early should I arrive on similar days?

Sensors and telemetry

Anonymous corridor footfall, doorway counts and lift usage at the relevant approaches; no identity or personal-device tracking. Verify the current route graph and the timestamp, health and coverage of each lift, door, closure and aggregate-footfall source used; do not treat an unobserved route segment as uncongested or accessible.

Authoritative supporting data

Authoritative current sources: seminar start times, room entrance and route topology, other scheduled class changes, temporary closures and current access rules. Verify current room use, access, bookability, equipment and services in authorised operational records; the May 2021 plan supplies spatial context only.

Analysis required

Historical and predictive analysis: measure queue and travel-time patterns before comparable sessions, identify recurring bottlenecks, forecast the next session and recommend an arrival buffer and alternative route.

Answer boundary

The system cannot determine whether an individual was late or change attendance policy. Counts may be uncertain at multi-door junctions, and recommendations must not route students through restricted areas. When history is sparse, sensing coverage has changed or future schedules are not comparable, widen the prediction interval or withhold the forecast.

Why this is useful: Helps students plan a reliable arrival time during tightly scheduled postgraduate teaching.

PGT-020

CORE

L2 - hybrid-session readiness

R2 - integration-dependent

Taught postgraduate question: Is today's room ready for a hybrid seminar, based on the room booking, AV and network status, access and background-noise evidence?

Sensors and telemetry

Required operational telemetry: authorised AV/conferencing controller state, display/input self-test, installed microphone or speaker endpoint status without ambient recording, network access-point/controller health, room/circuit power state and locally computed background LAeq where a calibrated acoustic point is correctly located. Use current inventory and time-stamped service-health feeds.

Authoritative supporting data

Authoritative current sources: confirmed booking and hybrid-use permission, current room equipment inventory and supported platform, IT/AV incident status, support availability, room access, accessibility requirements supplied by participants and an approved backup method. Historic room labels do not establish current hybrid capability.

Analysis required

Authoritative lookup + readiness assessment: map booking, access and every required hybrid function to current inventory, self-test/controller and incident status; distinguish room infrastructure from participant devices and external platforms; return ready, degraded, unavailable or unverified with the approved fallback.

Answer boundary

Room status cannot guarantee every laptop, account, remote connection, captioning service or speech intelligibility. Do not record ambient conversation. If an essential function lacks a current test or authoritative feed, label it unverified.

Why this is useful: Supports inclusive participation and reduces last-minute hybrid-teaching failures.

3. Independent study, dissertation and deadline peaks

Questions PGT-021-PGT-022

PGT-021

CORE

L3 - long-session recommendation

R2 - integration-dependent

Taught postgraduate question: When and where can I complete a four-hour dissertation-writing session with low noise, stable room conditions, power and working network service?

Sensors and telemetry

Room sound statistics, anonymous aggregate occupancy, temperature, CO2, illuminance, authorised aggregate network-service health and room electrical fault telemetry, with coverage and health metadata. Quality gate: use only commissioned or calibrated points with verified room or served-zone mapping, synchronised timestamps, current health and explicit missing-data flags. Corridor or adjacent-room readings are context only, never a room substitute.

Authoritative supporting data

Authoritative current sources: verified student-accessible spaces, bookings, opening hours, power-point inventory, eduroam status, timetable, route graph and weather forecast. Verify current room use, access, bookability, equipment and services in authorised operational records; the May 2021 plan supplies spatial context only.

Analysis required

Forecast plus recommendation: evaluate four-hour windows rather than snapshots, penalise spaces with high variability or frequent service faults, and rank a primary and backup option with criterion-specific confidence.

Answer boundary

Do not guarantee a desk, socket, personal thermal comfort or device-level network performance. If a room lacks continuous evidence across the full period, reduce confidence or omit that candidate. When history is sparse, sensing coverage has changed or future schedules are not comparable, widen the prediction interval or withhold the forecast.

Why this is useful: Supports sustained dissertation work where interruption and relocation are especially costly.

PGT-022

SITUATIONAL

L3 - daylight and glare recommendation

R3 - research-grade or restricted

Taught postgraduate question: Where can I work this afternoon with useful daylight but a low risk of distracting screen glare?

Sensors and telemetry

Required (proposed unless verified): desk- or task-plane illuminance at representative positions, window-facing luminance or calibrated glare proxy, solar radiation and blind state where available; room occupancy and current lighting state. Verify sensor geometry, orientation, calibration/commissioning and timestamp; a ceiling lux sensor alone cannot establish screen glare.

Authoritative supporting data

Authoritative current sources: verified student-accessible spaces, desk orientation and window geometry, blind-control availability, opening/booking status, route and a short-term solar/cloud forecast. Verify current room configuration operationally; the May 2021 plan is spatial context only.

Analysis required

Forecast + recommendation: estimate daylight and glare risk at candidate desk zones across the requested period using solar position, cloud forecast, window/blind state and measured illuminance; filter by access and availability; rank alternatives with uncertainty.

Answer boundary

This estimates environmental glare risk, not the student's visual condition or exact screen readability. Do not recommend a desk without valid task-position geometry. If blind state, orientation or calibrated glare evidence is absent, provide daylight observations only and mark glare not assessable.

Why this is useful: Helps a student use daylight productively without moving repeatedly when sunlight changes.

3. Independent study, dissertation and deadline peaks

Questions PGT-023-PGT-024

PGT-023

CORE

L3 - compute-study recommendation

R2 - integration-dependent

Taught postgraduate question: I need five hours for coding and data analysis. Which student space is most likely to offer stable power, working network service, low crowding and manageable heat?

Sensors and telemetry

Anonymous aggregate occupancy, room temperature, CO2, authorised aggregate network-service health, electrical fault telemetry and optional room or circuit demand for monitored candidates. Use time-aligned, authorised endpoint or controller telemetry and current inventory metadata; do not infer device readiness from room occupancy, and do not inspect content, credentials or identifiable network traffic.

Authoritative supporting data

Authoritative current sources: access and bookability, power provision, network service status, opening hours, bookings, room use and planned IT or electrical maintenance. Verify current room use, access, bookability, equipment and services in authorised operational records; the May 2021 plan supplies spatial context only.

Analysis required

Forecast and recommendation: risk forecast and recommendation: estimate the probability of interruption from crowding, service incidents and thermal instability using comparable long sessions; rank robust options and provide a backup with current recheck time.

Answer boundary

The answer cannot guarantee uninterrupted electricity, network performance or application availability and must not inspect the student's device or code. Unknown service telemetry must not be treated as healthy. When history is sparse, sensing coverage has changed or future schedules are not comparable, widen the prediction interval or withhold the forecast.

Why this is useful: Helps computational work whose progress may be lost or delayed by an avoidable interruption.

PGT-024

CORE

L3 - deadline-demand forecast

R2 - integration-dependent

Taught postgraduate question: My dissertation deadline is next week. On the busiest days, how early should I arrive for a reasonable chance of finding a suitable study space?

Sensors and telemetry

Historical anonymous aggregate occupancy, entrance counts, study-area footfall and room-level LAeq/L10/L90 sound statistics, with sensor coverage and room-status history. Quality gate: use only commissioned or calibrated points with verified room or served-zone mapping, synchronised timestamps, current health and explicit missing-data flags. Corridor or adjacent-room readings are context only, never a room substitute.

Authoritative supporting data

Authoritative current sources: dissertation and assessment calendar, timetable, bookings, opening hours, planned events and current access rules. Verify current room use, access, bookability, equipment and services in authorised operational records; the May 2021 plan supplies spatial context only.

Analysis required

Multi-day demand forecast + recommendation: define the required session lengths; model candidate-space demand for each day and arrival time using comparable deadline periods, bookings and teaching load; return two preferred sessions and backups with probability ranges.

Answer boundary

The forecast cannot guarantee a desk or reserve space. Treat changes in opening, room stock, access and sensing coverage as model breaks. If comparable deadline history is insufficient, provide verified opening and booking information without a demand probability.

Why this is useful: Helps students plan scarce time and avoid unnecessary stress during a concentrated deadline period.

3. Independent study, dissertation and deadline peaks

Questions PGT-025-PGT-026

PGT-025

SITUATIONAL

L2 - deadline-period comparison

R2 - integration-dependent

Taught postgraduate question: How do crowding, background noise and measured room conditions during this deadline week differ from ordinary teaching weeks?

Sensors and telemetry

Anonymous aggregate occupancy, corridor footfall, room-level LAeq/L10/L90 sound statistics, temperature and CO2 for the selected period and baseline weeks. Quality gate: use only commissioned or calibrated points with verified room or served-zone mapping, synchronised timestamps, current health and explicit missing-data flags. Corridor or adjacent-room readings are context only, never a room substitute.

Authoritative supporting data

Authoritative current sources: academic calendar, timetable, bookings, room-access changes, events and sensor deployment history. Verify current room use, access, bookability, equipment and services in authorised operational records; the May 2021 plan supplies spatial context only.

Analysis required

Historical comparison: define comparable weekdays and operating hours, calculate differences in typical and peak conditions, adjust for rooms that opened or closed, and report uncertainty and missing data by criterion.

Answer boundary

The result is building-level context and must not infer individual stress, productivity or attainment. A change in sensor coverage or room use must not be presented as a genuine behavioural change. If occupancy, weather, schedule, coverage or operating state are not comparable, label the comparison limited and do not assign a single cause.

Why this is useful: Makes deadline-period pressures visible and helps students choose alternative times or spaces.

PGT-026

SITUATIONAL

L4 - concentration-support assessment

R3 - research-grade or restricted

Taught postgraduate question: I am struggling to concentrate in this room. Are any measured conditions unusual, and is there a nearby student space that may suit focused work better?

Sensors and telemetry

Required telemetry (proposed unless verified): correctly located room temperature/relative humidity, NDIR CO2, PM2.5 where deployed, task illuminance, numerical LAeq/L10/L90 and anonymous aggregate occupancy; relevant HVAC state for context. Use verified in-room mapping, calibration or commissioning, current health, aligned clocks, sufficient coverage and explicit missingness.

Authoritative supporting data

Authoritative current sources: room and access status, bookings, route graph, permitted use and any preferences the student voluntarily supplies. Verify current room use, access, bookability, equipment and services in authorised operational records; the May 2021 plan supplies spatial context only.

Analysis required

Observation + contextual comparison + recommendation: compare current sound, occupancy, temperature, CO2 and lighting with this room's own time-matched baseline; identify which measured variables are unusual; rank nearby authorised alternatives using the student's stated study needs.

Answer boundary

The system cannot diagnose why the student cannot concentrate or make a health claim. A normal sensor reading does not invalidate the student's experience. Do not use speech or identity; when coverage is incomplete, state which factors were not assessed.

Why this is useful: Provides an evidence-based option to relocate without overstating what the building can know about concentration.

3. Independent study, dissertation and deadline peaks

Questions PGT-027-PGT-028

PGT-027

SITUATIONAL

L2 - restorative-break recommendation

R2 - integration-dependent

Taught postgraduate question: Where nearby can I take a short, relatively quiet break before returning to dissertation work?

Sensors and telemetry

Live anonymous aggregate occupancy, corridor footfall, privacy-preserving, locally computed sound-level statistics, temperature, CO2 and illuminance in verified break or waiting areas. Quality gate: use only commissioned or calibrated points with verified room or served-zone mapping, synchronised timestamps, current health and explicit missing-data flags. Corridor or adjacent-room readings are context only, never a room substitute.

Authoritative supporting data

Authoritative current sources: permitted use of spaces, current access, opening hours, bookings, route graph and temporary closures. Verify current room use, access, bookability, equipment and services in authorised operational records; the May 2021 plan supplies spatial context only.

Analysis required

Current-state lookup plus recommendation: find nearby spaces appropriate for a short break, rank lower-crowding and lower-sound options, estimate travel and return time, and retain a fallback.

Answer boundary

The building cannot judge whether a break is medically necessary or promise privacy. It must not recommend staff-only or booked spaces and must use only aggregate occupancy and sound statistics. Exclude every candidate with an unverified hard constraint; if no valid option remains, say so and provide the authorised next step rather than guess.

Why this is useful: Supports sustainable study habits during intensive postgraduate work without disrupting the student's schedule.

PGT-028

CORE

L1 - weekend access and status

R2 - integration-dependent

Taught postgraduate question: Which student spaces are open for dissertation work this weekend, and are any lifts, entrances or essential services unavailable?

Sensors and telemetry

Current room environmental status and anonymous aggregate occupancy where monitored; lift and door status only from authorised operational systems. Quality gate: use only commissioned or calibrated points with verified room or served-zone mapping, synchronised timestamps, current health and explicit missing-data flags. Corridor or adjacent-room readings are context only, never a room substitute.

Authoritative supporting data

Authoritative current sources: opening hours, access permissions, verified weekend space availability, bookings, planned works, lift and service status, security notices and support contacts. Verify current room use, access, bookability, equipment and services in authorised operational records; the May 2021 plan supplies spatial context only.

Analysis required

Authoritative lookup plus live context: list only verified accessible spaces and service limitations, distinguish official status from environmental observations, and provide a timestamp and contact route for unresolved information.

Answer boundary

The system must not infer access from an empty room or a sensor reading. Official access control, closures and security instructions take precedence, and the answer is not a personal-safety guarantee. Show the source and timestamp. If the feed is stale, unavailable or outside the student's permission, report 'unknown' rather than infer.

Why this is useful: Prevents an unnecessary journey and helps a student plan a weekend dissertation session responsibly.

3. Independent study, dissertation and deadline peaks

Questions PGT-029-PGT-030

PGT-029

SITUATIONAL

L3 - return-availability estimate

R2 - integration-dependent

Taught postgraduate question: If I leave for lunch for 45 minutes, what is the chance of finding a comparable study space when I return, and what backup should I use?

Sensors and telemetry

Live and historical anonymous aggregate occupancy, doorway counts, sound statistics and environmental conditions for comparable spaces. Quality gate: use only commissioned or calibrated points with verified room or served-zone mapping, synchronised timestamps, current health and explicit missing-data flags. Corridor or adjacent-room readings are context only, never a room substitute.

Authoritative supporting data

Authoritative current sources: access and permitted use, bookings, timetable, opening hours and the space requirements stated by the student. Verify current room use, access, bookability, equipment and services in authorised operational records; the May 2021 plan supplies spatial context only.

Analysis required

Forecast and recommendation: short-horizon estimate and fallback recommendation: forecast demand during the absence, identify equivalent spaces, provide a probability range and suggest a backup area and recheck time.

Answer boundary

The building cannot reserve a seat, watch belongings or track whether the student returns. It should not encourage leaving property unattended and must not reveal individual occupancy patterns. When history is sparse, sensing coverage has changed or future schedules are not comparable, widen the prediction interval or withhold the forecast.

Why this is useful: Helps the student decide whether to relocate materials or plan a more reliable return strategy.

PGT-030

CORE

L4 - evidence transparency

R1 - practical with core feeds

Taught postgraduate question: Why did you recommend this dissertation workspace? Please separate what you observed, looked up, calculated, estimated and forecast.

Sensors and telemetry

Sensor observations and their metadata: location, time, calibration or commissioning status, health, coverage and missingness; model input and output metadata for any forecast. Preserve the source identifier, observed property, unit, verified room or served-zone mapping, sample interval, latest timestamp, commissioning or calibration record, health state, coverage and missing-data flags for every item used.

Authoritative supporting data

Authoritative current sources: booking and access sources, room registry, network and service sources, route data, data-permission rules and model version. Verify current room use, access, bookability, equipment and services in authorised operational records; the May 2021 plan supplies spatial context only.

Analysis required

Provenance explanation: reconstruct the recommendation pipeline and label each claim as observation, authoritative lookup, calculation, estimate or forecast; list source location, period, quality checks, weighting, uncertainty, omitted criteria and fallback logic.

Answer boundary

Do not expose security-sensitive infrastructure, other users' data or hidden personal profiles. If the system cannot trace a claim to a permitted source and operation, withdraw that part of the recommendation rather than rationalise it after the fact.

Why this is useful: Lets a taught postgraduate judge whether a recommendation is evidence-based, current and suitable for the decision.

4. Specialist computer, laboratory and project work

Questions PGT-031-PGT-032

PGT-031

CORE

L3 - workstation and demand recommendation

R2 - integration-dependent

Taught postgraduate question: After 3 p.m., which computer lab am I allowed to use, and where am I most likely to find a workstation with the software I need?

Sensors and telemetry

Required operational telemetry: anonymous workstation availability or endpoint sign-in state only where privacy-approved and aggregated; workstation power/health, room occupancy, temperature and network-controller status. Do not inspect files, credentials, screen content or identifiable traffic; validate timestamps and endpoint inventory.

Authoritative supporting data

Authoritative current sources: student-accessible computer labs, current workstation inventory, required software/licence availability, access rights, teaching bookings, opening hours, maintenance/IT incidents and route status. Historic room labels are not evidence of current lab function or software provision.

Analysis required

Authoritative lookup + availability estimate + recommendation: filter labs by current access and software entitlement; estimate usable workstation availability from authorised aggregate endpoint status and demand history; rank options by service health, travel and room conditions with a backup.

Answer boundary

A low room count does not prove a compatible workstation is free, and endpoint status does not guarantee a successful personal login or licence checkout. If software, entitlement or endpoint telemetry is stale, report availability as unknown and direct the student to the official IT source.

Why this is useful: Saves time when specialist software and short practical windows make a failed lab visit costly.

PGT-032

CORE

L3 - service-risk forecast

R3 - research-grade or restricted

Taught postgraduate question: Where can I do a three-hour data-processing session with reliable power and network service?

Sensors and telemetry

Required telemetry: authorised network-controller and access-point health, upstream incident status, room/circuit power quality or outage state where instrumented, anonymous aggregate occupancy, room air temperature and relevant equipment-load context. Use time-aligned service and sensor streams with verified room/circuit mapping, current health and missing-data flags.

Authoritative supporting data

Authoritative current sources: current PGT-accessible rooms and permitted activity, bookings, opening hours, network and power incident feeds, socket provision, support arrangements and the student-specified processing needs. Specialist-room access and equipment must be verified operationally.

Analysis required

Forecast + recommendation: filter authorised spaces by access, power and network provision; combine current controller/circuit status with historical service incidents, occupancy, computing load and room temperature; rank a primary and backup for the requested three-hour window.

Answer boundary

Do not guarantee uninterrupted electricity, Wi-Fi, cloud service or the student's device. Whole-room or circuit evidence cannot prove a particular socket. If service topology or current status is unknown, omit that criterion or mark the room unverified.

Why this is useful: Supports computational coursework and analysis that may be disrupted by power, network or thermal instability.

4. Specialist computer, laboratory and project work

Questions PGT-033-PGT-034

PGT-033

SITUATIONAL

L4 - activity-suitability assessment

R3 - research-grade or restricted

Taught postgraduate question: I need to test a small hardware or sensing prototype for a module. Which specialist space is appropriate for the activity and its supervision, extraction, particulate-control and power needs?

Sensors and telemetry

Required evidence depends on the declared activity: authorised extraction airflow or local exhaust status, PM2.5/PM10 and indicative TVOC where relevant, room temperature, equipment/circuit power and door state; certified hazard systems remain separate. Use only risk-assessment-specified, correctly located and quality-controlled measurements.

Authoritative supporting data

Authoritative current sources: present specialist-room function, PGT authorisation, booking, induction, supervision, equipment inventory, activity-specific risk assessment, prohibited materials/processes, local rules and emergency procedures. The May 2021 room label cannot authorise the activity.

Analysis required

Authoritative suitability assessment: translate the stated activity into required room permissions, supervision, equipment, extraction, particulate control, bench space and electrical constraints; compare only currently authorised specialist spaces; return a compliant next step rather than a general room score.

Answer boundary

Environmental sensing cannot authorise an experiment or replace induction, risk assessment, supervision or equipment-specific procedures. Do not infer that a historic workshop or maker-room label is current. If the activity or competent approval is unclear, ask for clarification and return no room recommendation.

Why this is useful: Helps students use specialist facilities correctly without treating a generic room reading as safety approval.

PGT-034

SITUATIONAL

L3 - high-load lab forecast

R3 - research-grade or restricted

Taught postgraduate question: Which computer or project room is least likely to become too warm during a high-compute practical?

Sensors and telemetry

Required telemetry (proposed unless verified): room and representative workstation-inlet temperature, relative humidity, aggregate workstation or circuit power, cooling/HVAC airflow and supply temperature, fan or cooling-unit state and anonymous occupancy. Use time-synchronised streams with verified room/equipment/meter topology, current health, sufficient coverage and explicit gaps.

Authoritative supporting data

Authoritative current sources: current computer/project-room access, booking, workstation inventory, expected practical load, supported activities, cooling service relationships, maintenance incidents and outdoor forecast. Historic room use is not current evidence.

Analysis required

Forecast: identify currently authorised computing/project spaces; model room temperature and cooling margin for the requested practical from occupancy, expected electrical load, outdoor weather and HVAC state; back-test against comparable high-load sessions and rank options with intervals.

Answer boundary

The forecast concerns monitored room and equipment conditions, not processor performance or hardware safety. Without expected load, cooling state or representative inlet coverage, report current observations only and withhold the overheating ranking.

Why this is useful: Helps students and teaching staff avoid preventable disruption during compute-intensive practical work.

4. Specialist computer, laboratory and project work

Questions PGT-035-PGT-036

PGT-035

ADVANCED

L4 - lab heat diagnosis

R3 - research-grade or restricted

Taught postgraduate question: Why was the computer lab hotter than usual during yesterday's practical?

Sensors and telemetry

Required telemetry: distributed room and representative workstation-inlet temperatures, aggregate workstation/circuit power, anonymous occupancy, supply airflow and temperature, cooling/fan state, door state and outdoor weather. Use verified room, circuit and HVAC relationships, synchronised clocks, quality flags and comparable coverage before and during the practical.

Authoritative supporting data

Authoritative current sources: practical-session timeline, room and HVAC topology, equipment schedule, setpoints, faults, overrides and maintenance events. Verify current room use, access, bookability, equipment and services in authorised operational records; the May 2021 plan supplies spatial context only.

Analysis required

Diagnosis: align room temperature, occupancy, plug-load, outdoor weather, solar gain and HVAC command/response around the practical; compare with matched sessions; rank supported explanations and show unmeasured variables and conflicting evidence.

Answer boundary

A temporal association is not proof of cause. Do not assign blame to students, equipment or facilities without corresponding evidence. If load, airflow or radiant-gain data are absent, report the observed sequence and plausible alternatives only.

Why this is useful: Supports targeted facilities or IT follow-up instead of attributing the problem vaguely to a busy room.

PGT-036

CORE

L1 - authorised-use lookup

R2 - integration-dependent

Taught postgraduate question: Can I use this specialist room today, and what booking, induction, supervision or risk-assessment conditions apply?

Sensors and telemetry

Sensors are secondary: current room access or door status, extraction or equipment-health telemetry and room environmental status may be shown only after authorisation is established. Use authorised, time-stamped controller, inventory or incident feeds with current service-health metadata. A missing or stale operational feed is 'unknown' and cannot be inferred from environmental sensors.

Authoritative supporting data

Authoritative current sources: present room function, student-role access, booking, induction completion, supervision requirement, project risk assessment, equipment permissions, opening hours and responsible technical contact. Current specialist-room purpose, access, equipment and supervision must be verified operationally; the May 2021 drawing is not authority for present use.

Analysis required

Authoritative permission lookup: verify each prerequisite and return authorised, conditionally authorised, not authorised or unknown, with the unmet requirement and official next step. Do not use occupancy or an unlocked door as evidence of permission.

Answer boundary

The system cannot grant access, waive training or expose another person's credentials. If a prerequisite source is stale, unavailable or outside the student's permission, return unknown or not authorised and direct the student to responsible staff.

Why this is useful: Prevents accidental misuse and gives the student a clear route to satisfy the required conditions.

4. Specialist computer, laboratory and project work

Questions PGT-037-PGT-038

PGT-037

CORE

L3 - remote-compute session recommendation

R2 - integration-dependent

Taught postgraduate question: I need to monitor a long remote GPU or cloud job. Where can I work with reliable room power and network service, low background noise and enough access time before closing?

Sensors and telemetry

Required telemetry: authorised room/circuit power status, network access-point/controller and upstream-service health, room temperature and anonymous aggregate occupancy or numerical background sound for candidate work areas. The remote GPU/cloud platform status must come from its own authorised service, not building sensors. Require verified room mapping, synchronised timestamps, health and gap flags; corridor data is contextual and network evidence remains aggregate.

Authoritative supporting data

Authoritative current sources: PGT-accessible spaces, permitted online-call/compute activity, opening and exit times, bookings, network incidents, planned power work, external platform status if available, and only the job-duration requirements the student supplies.

Analysis required

Authoritative lookup + forecast + recommendation: filter spaces by permitted duration, closing time, power and network provision; check current service status and forecast crowding, sound and room temperature; leave enough time for saving/checkpointing and an authorised exit.

Answer boundary

The building cannot monitor or guarantee the remote job, cloud platform, account, laptop battery or data integrity. Do not inspect traffic or workload content. If the job requires unattended equipment, apply the relevant institutional policy rather than recommending an ordinary study space.

Why this is useful: Supports reliable completion of computational coursework without a late-session access or service surprise.

PGT-038

CORE

L2 - project-demo room recommendation

R2 - integration-dependent

Taught postgraduate question: Where should we test our project demonstration if we need a display, power, network service, low background noise and space for equipment?

Sensors and telemetry

Room sound statistics, occupancy, temperature, CO2, authorised aggregate network-service health and authorised AV and electrical status. Use time-aligned, authorised endpoint or controller telemetry and current inventory metadata; do not infer device readiness from room occupancy, and do not inspect content, credentials or identifiable network traffic.

Authoritative supporting data

Authoritative current sources: room capacity and use, equipment and power inventory, booking, access, permitted project activity, route constraints and support status. Verify current room use, access, bookability, equipment and services in authorised operational records; the May 2021 plan supplies spatial context only.

Analysis required

Comparison and recommendation: apply hard equipment, access and capacity requirements; compare live conditions and service state; rank authorised rooms and state which criteria are confirmed, historical or unknown.

Answer boundary

Do not infer that a room permits equipment setup merely because it is large or empty. AV and power readiness must come from authorised systems and the result cannot create a booking. If occupancy, weather, schedule, coverage or operating state are not comparable, label the comparison limited and do not assign a single cause.

Why this is useful: Reduces setup risk before a project demonstration or assessment rehearsal.

4. Specialist computer, laboratory and project work

Questions PGT-039-PGT-040

PGT-039

SITUATIONAL

L4 - service-incident reconstruction

R3 - research-grade or restricted

Taught postgraduate question: Power or network service failed during my lab session. Does the evidence point to a room-level fault, a wider building incident or an external service problem?

Sensors and telemetry

Required telemetry: time-stamped room/circuit voltage, breaker, UPS or power-quality events where available; network access-point/controller, switch and upstream incident status; room equipment-controller alarms and neighbouring-zone service state. Use authorised logs with synchronised clocks, verified topology and retention adequate for the incident window.

Authoritative supporting data

Authoritative current sources: exact incident time supplied by the student, current room/circuit/network topology, planned maintenance, IT and estates incident records, affected-service scope and external-provider status. Do not use the historic floor plan as current infrastructure topology.

Analysis required

Incident reconstruction and diagnosis: align all logs around the failure; determine whether loss was local, floor/building-wide or external; compare command, state and outage events; rank the supported fault domain and identify the evidence needed for a more specific cause.

Answer boundary

Absence of a logged event is not proof that service did not fail, particularly when telemetry shares the failed infrastructure. Do not expose network-security details or personal traffic. If clocks or logs are incomplete, report the observed scope and unresolved hypotheses only.

Why this is useful: Helps the student contact the correct support service and document an assessed-work disruption accurately.

PGT-040

SITUATIONAL

L3 - specialist-space demand forecast

R2 - integration-dependent

Taught postgraduate question: When is the specialist lab usually least busy, and which available slot this week is realistic for my work?

Sensors and telemetry

Required telemetry: authorised booking and aggregate workstation/session availability, anonymous doorway counts or occupancy, room-condition telemetry and equipment-service state. Use current inventory, source timestamps and validated count coverage; do not identify users or infer activity from personal devices. Use aggregate service or endpoint health only; do not inspect content, credentials or identifiable traffic.

Authoritative supporting data

Authoritative current sources: present specialist-lab purpose, PGT eligibility, induction/supervision conditions, bookings, teaching timetable, opening hours, equipment maintenance and this week's permitted slots. The May 2021 room label is not proof of current specialist use.

Analysis required

Historical demand estimate + authorised-slot recommendation: model aggregate use by comparable weekday/time while accounting for bookings and teaching; filter by the student's access, induction and activity requirements; rank slots with uncertainty and a current recheck.

Answer boundary

Do not reveal named users, sensitive laboratory schedules or security detail. A low aggregate count does not prove equipment availability or permission. If access, booking or occupancy coverage is incomplete, return the official booking route rather than a demand forecast.

Why this is useful: Makes limited specialist access easier to plan around a taught postgraduate timetable.

5. Group projects, meetings and presentations

Questions PGT-041-PGT-042

PGT-041

CORE

L3 - group-space recommendation

R2 - integration-dependent

Taught postgraduate question: Six of us need two hours for a group project. Where can we talk, use a display and power, and avoid excessive crowding or poor ventilation?

Sensors and telemetry

Required telemetry (proposed unless verified): correctly located room temperature/relative humidity and NDIR CO2, numerical LAeq/L10/L90, anonymous aggregate occupancy, room/circuit power status and authorised AV-controller state for candidate group rooms. Use calibrated or commissioned points, current health, room mapping and missing-data flags.

Authoritative supporting data

Authoritative current sources: current PGT-accessible and discussion-permitted rooms, booking availability, capacity, display/power inventory, opening hours, accessibility, route and planned events. The May 2021 plan is historical geometry only.

Analysis required

Recommendation: apply hard constraints for six people, discussion permission, booking, power and display; forecast crowding, sound and CO2 for the two-hour window from comparable periods; rank a primary and backup with the evidence time and uncertainty.

Answer boundary

An occupancy count does not prove a room is bookable or that seats, sockets or AV are free. Do not treat low sound as privacy. Exclude any candidate with unverified access, capacity or equipment and say when no valid option remains.

Why this is useful: Helps the group choose a room that supports collaboration without displacing or disturbing other users.

PGT-042

CORE

L3 - inclusive hybrid-room recommendation

R2 - integration-dependent

Taught postgraduate question: We need a room for eight people and one remote participant. Which available option best meets step-free access, hybrid-equipment and room-condition needs?

Sensors and telemetry

Design requirement: official hybrid-room AV/conferencing and aggregate network status, room/circuit power, in-room temperature and direct NDIR CO2, numerical background sound, anonymous occupancy and current lift/door status. No ambient speech recording. Require verified room/route mapping, time alignment, current health and gap flags.

Authoritative supporting data

Authoritative current sources: current room capacity and eight-person-plus-remote suitability, booking, PGT access, supported hybrid platform, step-free route and door/lift status, accessibility features supplied by participants, IT/AV incidents and support fallback.

Analysis required

Forecast and recommendation: inclusive recommendation: treat capacity, step-free access and required hybrid functions as hard constraints; classify service readiness; forecast room conditions for the meeting period; rank valid options and provide an accessible backup independent of the same failed service where possible.

Answer boundary

Do not infer disability, share accessibility preferences without consent or guarantee remote connectivity. Numerical sound does not establish speech intelligibility. If the step-free route or an essential hybrid function is unverified, do not recommend the room.

Why this is useful: Supports equitable group participation without forcing members to disclose more personal information than necessary.

5. Group projects, meetings and presentations

Questions PGT-043-PGT-044

PGT-043 CORE L3 - rehearsal-space recommendation R2 - integration-dependent

Taught postgraduate question: Where can we rehearse our presentation for 90 minutes with a working display, controllable lighting, low background noise and reliable network service?

Sensors and telemetry

Required telemetry: authorised display/input and room-controller state, network access-point/controller health, room/circuit power, task illuminance and controllable-lighting state, locally computed numerical sound statistics and anonymous occupancy. Use current inventory, source timestamps and sensor placement/health metadata; record no speech.

Authoritative supporting data

Authoritative current sources: current rehearsal-permitted rooms, booking, capacity, display and connection inventory, lighting controls, access and opening hours, presentation time and any accessibility requirements supplied by the group.

Analysis required

Lookup plus recommendation: verify room and service readiness for the 90-minute period; compare numerical background sound and lighting range; filter by booking/access; rank a primary and backup and identify what the group must test on arrival.

Answer boundary

Operational status cannot guarantee a particular laptop, cable, file, account, speech intelligibility or privacy. The system must not capture rehearsal content. If AV or booking status is unverified, direct the group to an authorised test/support route rather than claim readiness.

Why this is useful: Provides a realistic rehearsal environment and reduces last-minute technical surprises.

PGT-044 CORE L3 - deadline-week group forecast R2 - integration-dependent

Taught postgraduate question: During group-project deadline week, which time windows are most likely to offer a suitable meeting space for our team?

Sensors and telemetry

Historical anonymous aggregate occupancy, doorway counts and sound statistics in verified group spaces. Quality gate: use only commissioned or calibrated points with verified room or served-zone mapping, synchronised timestamps, current health and explicit missing-data flags. Corridor or adjacent-room readings are context only, never a room substitute.

Authoritative supporting data

Authoritative current sources: project and assessment calendar, bookings, timetable, room capacities, permitted use, opening hours and current access. Verify current room use, access, bookability, equipment and services in authorised operational records; the May 2021 plan supplies spatial context only.

Analysis required

Forecast and recommendation: demand forecast: compare previous deadline periods and current bookings, estimate availability by day and time, and return several robust options with uncertainty and a recommended booking lead time.

Answer boundary

The forecast cannot reserve a room or infer which groups are present. Changed room access or deadline schedules must be incorporated rather than assuming past patterns will repeat. When history is sparse, sensing coverage has changed or future schedules are not comparable, widen the prediction interval or withhold the forecast.

Why this is useful: Helps a group coordinate early during a compressed postgraduate assessment period.

5. Group projects, meetings and presentations

Questions PGT-045-PGT-046

PGT-045

SITUATIONAL

L4 - presentation-room comparison

R2 - integration-dependent

Taught postgraduate question: Which of these two rooms is the better choice for an assessed presentation when capacity, accessibility, acoustics, lighting, network service and room conditions are considered together?

Sensors and telemetry

Numerical acoustic metrics, illuminance, temperature, CO2 and occupancy; authorised network, AV and lift status. Use time-aligned, authorised endpoint or controller telemetry and current inventory metadata; do not infer device readiness from room occupancy, and do not inspect content, credentials or identifiable network traffic. Require verified room and route mapping, synchronised timestamps, health and gap flags; corridor data is contextual, unobserved route segments are unknown, and network evidence remains aggregate.

Authoritative supporting data

Authoritative current sources: room capacity, layout, accessibility, booking, equipment inventory, assessment requirements, route graph and support status. Verify current room use, access, bookability, equipment and services in authorised operational records; the May 2021 plan supplies spatial context only.

Analysis required

Multi-criteria comparison: apply assessment and accessibility constraints, compare confirmed equipment and live conditions, use historical stability for the expected time, and explain the trade-offs rather than producing an unexplained score.

Answer boundary

Sensor data cannot assess presentation quality or guarantee speech intelligibility for every listener. Official room allocation and assessment rules override the recommendation. If occupancy, weather, schedule, coverage or operating state are not comparable, label the comparison limited and do not assign a single cause.

Why this is useful: Supports a transparent room choice for a high-stakes group presentation.

PGT-046

SITUATIONAL

L4 - meeting-sequence optimisation

R2 - integration-dependent

Taught postgraduate question: Our group has two meetings and a seminar on the same day. Can you plan an accessible sequence of rooms with enough transfer time and less exposure to peak corridor congestion?

Sensors and telemetry

Corridor and stair footfall, lift status and anonymous aggregate occupancy in candidate rooms; room environmental conditions for longer meetings. Verify the current route graph and the timestamp, health and coverage of each lift, door, closure and aggregate-footfall source used; do not treat an unobserved route segment as uncongested or accessible.

Authoritative supporting data

Authoritative current sources: agreed meeting windows supplied by the group, seminar location, bookings, access, room capacities, route graph, accessibility requirements and temporary closures. Verify current room use, access, bookability, equipment and services in authorised operational records; the May 2021 plan supplies spatial context only.

Analysis required

Forecast: itinerary optimisation: evaluate valid room sequences, forecast class-change congestion, preserve transfer buffers and hard accessibility constraints, and rank options with a fallback sequence.

Answer boundary

Use only schedule information the group chooses to share and do not track members. The system cannot create bookings or route users through restricted areas, and all live service states require recheck. When history is sparse, sensing coverage has changed or future schedules are not comparable, widen the prediction interval or withhold the forecast.

Why this is useful: Reduces coordination overhead and makes a complex group day more feasible.

5. Group projects, meetings and presentations

Questions PGT-047-PGT-048

PGT-047

CORE

L3 - move-or-break recommendation

R3 - research-grade or restricted

Taught postgraduate question: Our group room is getting warm and stuffy. Should we ask for a break, move to another suitable space, or report a facilities problem?

Sensors and telemetry

Required telemetry (proposed unless verified): correctly located in-room air temperature/relative humidity, NDIR CO2, anonymous aggregate occupancy and serving-zone airflow/fan state; optional PM2.5 or indicative TVOC only where installed and quality-controlled. Use current room/HVAC mapping, synchronised timestamps and explicit missingness.

Authoritative supporting data

Authoritative current sources: current booking and organiser, group size, meeting duration, authorised nearby alternatives, room access, facilities reporting route and any pre-agreed operational bands. Historic room labels do not establish current use or alternatives.

Analysis required

Observation + bounded diagnosis + recommendation: verify current in-room CO2, temperature and occupancy; check ventilation status and persistence; compare with the room's normal occupied pattern; rank three actions - continue and recheck, ask the organiser for a break/move, or report facilities - using explicit triggers.

Answer boundary

The system advises; the organiser controls the session and facilities staff diagnose equipment. CO2 and temperature do not establish individual health or complete air quality. If the readings are stale, spatially invalid or conflicting, direct the group to the organiser and official reporting route.

Why this is useful: Provides a proportionate response to a common group-work problem while respecting building controls.

PGT-048

SITUATIONAL

L2 - space-level disturbance assessment

R3 - research-grade or restricted

Taught postgraduate question: While our group was meeting, were sound levels in the nearby designated quiet-study areas unusually high compared with their normal pattern?

Sensors and telemetry

Required telemetry (proposed unless verified): fixed numerical LAeq/L10/L90 sound points inside the designated quiet-study areas and, where needed, a separate point in the group room; anonymous aggregate occupancy and door state for context. Use calibrated placement, synchronised clocks, sufficient coverage and no raw audio or speech features.

Authoritative supporting data

Authoritative current sources: current designation and permitted use of the quiet-study areas, group-room booking and meeting time, door relationships and any published acoustic expectations. The May 2021 plan does not prove current quiet-area designation.

Analysis required

Historical space-level comparison: align the group-meeting interval with numerical sound statistics in designated quiet areas; compare against time-matched baselines and nearby footfall; report whether levels were unusual and how long the deviation persisted.

Answer boundary

Do not record speech, identify a group, infer conversation content or claim that one meeting caused the sound increase. If acoustic points are not inside the quiet areas or coverage is incomplete, the disturbance question is not assessable.

Why this is useful: Helps groups choose and use collaborative spaces more considerately.

5. Group projects, meetings and presentations

Questions PGT-049-PGT-050

PGT-049

CORE

L3 - large-group suitability forecast

R3 - research-grade or restricted

Taught postgraduate question: We need a room for a 12-person project workshop. Which available option has enough capacity and the lowest predicted risk of ventilation or noise problems?

Sensors and telemetry

Required telemetry (proposed unless verified): distributed or demonstrably representative in-room NDIR CO2, air temperature/relative humidity and numerical sound statistics; anonymous aggregate occupancy; serving-HVAC airflow or fan state; authorised AV/network/power status if required. Use verified room/HVAC mapping, calibration, spatial coverage and missing-data flags.

Authoritative supporting data

Authoritative current sources: current rooms authorised for a 12-person workshop, capacity, booking, furniture/activity rules, access and step-free route, equipment requirements, opening hours and scheduled neighbouring activity. The historic drawing is not current authority.

Analysis required

Forecast and recommendation: filter by 12-person capacity, activity, access and services; forecast CO2, temperature and background sound for the requested duration using comparable group sessions and current HVAC state; rank valid options with prediction intervals and a backup.

Answer boundary

A single sensor cannot establish uniform conditions in a large or irregular room. The forecast cannot guarantee comfort, quietness or a booking. If capacity, room-level coverage or HVAC context is unverified, omit that candidate rather than infer suitability.

Why this is useful: Supports a workshop format that is more intensive than an ordinary meeting.

PGT-050

SITUATIONAL

L4 - matched pre/post comparison

R2 - integration-dependent

Taught postgraduate question: We used a different room for our meeting last week. Did background noise, crowding or ventilation indicators improve under comparable conditions?

Sensors and telemetry

Room sound statistics, occupancy, direct CO2, temperature and corridor footfall for both sessions, with coverage and sensor-health metadata. Quality gate: use only commissioned or calibrated points with verified room or served-zone mapping, synchronised timestamps, current health and explicit missing-data flags. Corridor or adjacent-room readings are context only, never a room substitute.

Authoritative supporting data

Authoritative current sources: meeting times, group size supplied voluntarily, room bookings, permitted use, room characteristics and any concurrent events. Verify current room use, access, bookability, equipment and services in authorised operational records; the May 2021 plan supplies spatial context only.

Analysis required

Matched pre/post comparison: align session duration and attendance, control for timetable and outdoor conditions where relevant, quantify changes by criterion and state whether differences exceed normal variability.

Answer boundary

A single pair of sessions cannot establish that the room caused the change. Do not identify participants or interpret CO2 as complete air quality, and disclose any unmatched conditions. If occupancy, weather, schedule, coverage or operating state are not comparable, label the comparison limited and do not assign a single cause.

Why this is useful: Helps a project group learn which space characteristics genuinely support its working style.

6. Part-time, commuting and international-student routines

Questions PGT-051-PGT-052

PGT-051

CORE

L2 - pre-travel status check

R2 - integration-dependent

Taught postgraduate question: Before I set off on a long commute, can you check whether my session, room access, lift and essential services are still available, and suggest somewhere to study beforehand?

Sensors and telemetry

Required operational evidence: official room-change, opening, access, lift, entrance, network and essential-service incident feeds; for a study-space suggestion, anonymous occupancy and correctly located room conditions. Use current timestamps and service-health metadata; do not infer class status from sensors or inspect identifiable network activity.

Authoritative supporting data

Authoritative current sources: official module/session status, current room and time, PGT access, lift/entrance/closure status, IT service incidents, opening hours, route graph, authorised study spaces and only the visit details the student submits for this check.

Analysis required

Authoritative pre-travel lookup + recommendation: verify the session, room, access, lift/entrance and essential service feeds for the planned window; classify each dependency as confirmed, degraded, unavailable or unknown; recommend a study space only from currently authorised options and state when to recheck.

Answer boundary

This cannot guarantee that teaching will proceed, excuse absence or expose security-sensitive status. Retain only the minimum visit details required for the check. If an official feed is stale or unavailable, show 'unknown' and direct the student to the named official source.

Why this is useful: Reduces the cost of an unnecessary or poorly prepared commute.

PGT-052

SITUATIONAL

L2 - route and policy lookup

R2 - integration-dependent

Taught postgraduate question: My train is delayed. What is the fastest open route to the seminar, and where can I check the official late-arrival procedure?

Sensors and telemetry

Required operational sources: current entrance, lift, door and closure state and aggregate route footfall where privacy-preserving counters are deployed. Environmental sensors are not evidence of route authorisation or academic policy. Verify route topology, timestamps, coverage and service health. Use current official status and instrumented route segments; an unobserved segment remains unknown.

Authoritative supporting data

Authoritative current sources: seminar room and start time supplied by the student, current authorised route graph, access permissions, temporary closures, step-free requirements if requested, and official module or School late-arrival guidance and contact.

Analysis required

Lookup plus travel-time estimate: compute the fastest currently authorised route from the stated arrival point; use aggregate footfall only to estimate delay; present a step-free alternative when requested; separately quote or link the official late-arrival procedure.

Answer boundary

The building cannot excuse lateness, guarantee admission or notify teaching staff unless an authorised workflow exists. Do not track the student after the query. If route or policy data is stale, provide the official contact and mark the estimate unavailable.

Why this is useful: Helps a commuter reach class efficiently while keeping academic decisions with the correct people.

6. Part-time, commuting and international-student routines

Questions PGT-053-PGT-054

PGT-053

CORE

L3 - evening and weekend recommendation

R2 - integration-dependent

Taught postgraduate question: Which evening or weekend study times are most dependable for access, network service, low crowding and stable measured conditions?

Sensors and telemetry

Required evidence: official entrance, lift, access and aggregate network status; anonymous occupancy and doorway counts; in-room temperature, direct NDIR CO2 and numerical sound for candidate study spaces. Require current service timestamps, verified room/route mapping, sensor health and gap flags; no identity or device tracking.

Authoritative supporting data

Authoritative current sources: actual PGT-accessible evening/weekend spaces, opening and staffed-support hours, bookings, closures, transport or exit constraints where supplied, network incidents and planned maintenance. Current use and access must not be inferred from the May 2021 plan.

Analysis required

Historical comparison and recommendation: compare equivalent evening/weekend windows for access incidents, aggregate demand, network degradation and room-condition variability; apply current opening and route constraints; rank reliable windows with sample size, uncertainty and a same-day check.

Answer boundary

This is not a personal-safety assessment or a guarantee of access, Wi-Fi, a seat or support. Do not expose exact after-hours occupancy. If opening, access or a critical service is unverified, do not recommend that window.

Why this is useful: Supports students who combine postgraduate study with employment or caring responsibilities.

PGT-054

CORE

L4 - one-day campus optimisation

R2 - integration-dependent

Taught postgraduate question: I am on campus for only one day this week. Can you plan a practical sequence for study, an online call, printing and a group meeting around my classes?

Sensors and telemetry

Room occupancy, footfall, sound statistics, temperature, CO2 and authorised aggregate network-service health; printer, AV and lift status through authorised service systems. Verify the current route graph and the timestamp, health and coverage of each lift, door, closure and aggregate-footfall source used; do not treat an unobserved route segment as uncongested or accessible.

Authoritative supporting data

Authoritative current sources: personal timetable supplied with consent, bookings, room features and access, printing-service status, opening hours, route graph, group meeting window and temporary closures. Verify current room use, access, bookability, equipment and services in authorised operational records; the May 2021 plan supplies spatial context only.

Analysis required

Itinerary optimisation: treat classes and required services as fixed constraints; schedule study, call, printing and meeting windows using current opening/access, bookings, service status, routes and forecast crowding; minimise travel and include time-feasible backups.

Answer boundary

Do not infer the student's attendance or retain a full personal schedule beyond the request. Printer queues, Wi-Fi and room availability remain estimates unless confirmed by authoritative status. If a service feed is unknown, offer the official check route and avoid a brittle itinerary.

Why this is useful: Makes a limited campus day more productive for part-time and commuting students.

6. Part-time, commuting and international-student routines

Questions PGT-055-PGT-056

PGT-055

CORE

L2 - arrival-point route recommendation

R2 - integration-dependent

Taught postgraduate question: From the arrival point I provide, which open entrance and step-free route should I use to reach my room?

Sensors and telemetry

Required operational sources: current entrance, door, lift and closure state; aggregate route footfall only when useful for delay context. Environmental sensors do not establish access. Verify every route segment, source timestamp and service-health state. Use current official status and instrumented route segments; an unobserved segment remains unknown.

Analysis required

Lookup and route recommendation: remove closed, restricted or non-step-free segments; calculate feasible routes and travel time; rank by the student's stated priority and provide a verified fallback and source time.

Authoritative supporting data

Authoritative current sources: arrival point supplied for the query, destination, current authorised and step-free route graph, open entrances, access permissions, temporary closures, lift status and official accessibility assistance contact.

Answer boundary

Do not infer or retain the student's location beyond the request. A route is not step-free unless every required segment and lift is verified. Official signage, staff directions and emergency closures override the recommendation.

Why this is useful: Reduces uncertainty for students who visit the building less frequently.

PGT-056

CORE

L1 - plain-language wayfinding

R1 - practical with core feeds

Taught postgraduate question: I am new to Abacws. Can you explain how the floors and room numbers work and guide me to my seminar using clear landmarks?

Sensors and telemetry

No environmental sensor is required for naming or basic wayfinding. Optional authorised lift, entrance, door and closure status can make the route current; aggregate footfall may support a broad congestion estimate. Use a verified current topology and time-stamped operational status. Use current official status and instrumented route segments; an unobserved segment remains unknown.

Analysis required

Authoritative lookup + plain-language route generation: resolve the current destination, floor, entrance and permitted route; translate verified room/floor identifiers into clear landmarks and turn-by-turn instructions; include lift or step-free details only when current status is known.

Authoritative supporting data

Authoritative current sources: current room and floor naming, public/student route graph, destination, access permissions, open entrances, lift and closure status, verified landmarks and plain-language accessibility information. The May 2021 drawing is a historical aid only.

Answer boundary

Do not rely on a historic room label or invent landmarks. Route guidance must respect access restrictions and temporary closures. If the destination or current route graph is ambiguous, ask for clarification or direct the student to official signage/reception.

Why this is useful: Helps international and other new taught postgraduates navigate without needing to understand building conventions.

6. Part-time, commuting and international-student routines

Questions PGT-057-PGT-058

PGT-057

SITUATIONAL

L3 - low-crowding route recommendation

R2 - integration-dependent

Taught postgraduate question: I would rather walk a little farther than use a very crowded corridor. Which open route should I take now?

Sensors and telemetry

Required evidence: privacy-preserving aggregate corridor/stair footfall and doorway counts at relevant route segments; current lift, door and closure status. Use verified route mapping, aligned timestamps, known counter coverage and no identity or personal-device tracking. Use current official status and instrumented route segments; an unobserved segment remains unknown.

Authoritative supporting data

Authoritative current sources: destination and current authorised route graph, access permissions, temporary closures, step-free requirement if any, and the student's stated willingness to trade distance for lower crowding.

Analysis required

Current estimate and recommendation: estimate segment congestion from recent counts, calculate travel time for feasible routes and rank the Pareto trade-off between distance and crowding; provide a fallback and recheck time around class change.

Answer boundary

An uninstrumented segment is unknown, not empty. Counts may be inaccurate at multi-entry junctions and do not identify people. Never route through restricted areas; if coverage is insufficient, provide the shortest verified route without a crowding claim.

Why this is useful: Supports confidence and orientation for students who are unfamiliar with the building or prefer lower-crowding routes.

PGT-058

CORE

L2 - pre-evening-call recommendation

R2 - integration-dependent

Taught postgraduate question: Where can I join a 30-minute online call before my evening class, while still being able to reach the class and leave the building afterwards?

Sensors and telemetry

Required telemetry: numerical LAeq/L10/L90 inside candidate rooms, anonymous occupancy, authorised network-service health, room/circuit power status and door state; optional room temperature/NDIR CO2 for a longer wait. Record no speech and use current sensor/service-quality metadata. Require verified room mapping, synchronised timestamps, health and gap flags; corridor data is contextual and network evidence remains aggregate.

Authoritative supporting data

Authoritative current sources: rooms currently authorised for student online calls, booking or drop-in rules, opening and exit times, next-class room and access route, network incidents, capacity and accessibility. The May 2021 plan does not establish present call suitability.

Analysis required

Lookup plus recommendation: filter to authorised call-suitable spaces available for the full interval and a viable route to class; classify network/power status; compare aggregate sound and occupancy; rank a primary and backup and state the latest departure time.

Answer boundary

Low background sound does not guarantee conversational privacy or speech intelligibility. The system must not listen to the call or guarantee the external meeting platform. If no enclosed or appropriately designated space is verified, say so and provide the official alternative.

Why this is useful: Supports a common pattern for part-time students balancing work calls and evening teaching.

6. Part-time, commuting and international-student routines

Questions PGT-059-PGT-060

PGT-059

SITUATIONAL

L4 - attend-or-remote decision support

R3 - research-grade or restricted

Taught postgraduate question: Is it worth travelling to Abacws for an optional study day tomorrow? What study spaces, services and likely conditions are available, so I can compare them with working remotely?

Sensors and telemetry

For Abacws only: proposed room-level anonymous occupancy, numerical sound, air temperature/relative humidity, direct NDIR CO2 and task illuminance; authorised network and room/circuit power status. Use verified in-room mapping, source timestamps, health, coverage and missing-data flags. Require verified room mapping, synchronised timestamps, health and gap flags; corridor data is contextual and network evidence remains aggregate.

Authoritative supporting data

Authoritative current sources: actual PGT-accessible spaces, bookings, opening hours, network/power incidents, timetable, route and weather. Remote-study conditions, travel burden and personal preferences are used only when the student explicitly supplies them for this comparison.

Analysis required

Decision support: verify tomorrow's opening, access, bookings and service status; forecast candidate-space crowding and measured conditions; compare only with the remote-study criteria the student chooses to provide; present trade-offs rather than a single prescriptive answer.

Answer boundary

Do not infer home conditions, costs, caring duties, disability or attendance obligations. The system cannot decide whether travel is personally worthwhile. If campus or remote evidence is incomplete, show the unknowns and leave the decision to the student.

Why this is useful: Helps a commuter make a transparent choice about an optional campus visit.

PGT-060

CORE

L2 - opt-in disruption notification

R2 - integration-dependent

Taught postgraduate question: Can I opt in to an alert if a room change, lift problem, entrance closure, network incident or other confirmed service issue affects my planned visit?

Sensors and telemetry

No continuous personal sensing is required. Use only authorised event feeds for official room changes, lift/entrance closures, access-service incidents, network or power outages and, where configured, room-level environmental alerts. Require source identity, event time, severity, expiry and service-health metadata. Use current official route and aggregate service status; unobserved segments remain unknown and no content, credentials or identifiable traffic may be inspected.

Authoritative supporting data

Authoritative current sources: an opt-in visit plan with the minimum times/locations needed, official timetable and room-change notifications, building and IT incidents, notification preferences, consent, expiry and revocation state.

Analysis required

Event matching + opt-in notification: monitor only the submitted visit window and authorised dependencies; deduplicate and prioritise official events; determine whether each event affects the planned location/time; send the source, timestamp, expected impact, official action and expiry.

Answer boundary

Do not infer attendance, continuously track the student or retain the visit plan beyond its purpose. Notifications are advisory and may be delayed; official course and emergency communications take precedence. The student must be able to pause, amend and revoke the service.

Why this is useful: Gives part-time and commuting students time to adjust before travelling.

7. Accessibility, well-being, navigation and live conditions

Questions PGT-061-PGT-062

PGT-061

CORE

L3 - accessible low-stimulation routing

R2 - integration-dependent

Taught postgraduate question: What is the best step-free route to my seminar if I also prefer lower crowding and lower background noise?

Sensors and telemetry

Required operational evidence: current lift, entrance, accessible-door and closure state; privacy-preserving aggregate corridor/stair footfall; numerical sound statistics only on route segments where correctly deployed. Verify every route segment, source timestamp and coverage; record no speech or identity. Require verified room and route mapping, synchronised timestamps, health and gap flags; corridor data is contextual and unobserved route segments remain unknown.

Authoritative supporting data

Authoritative current sources: verified step-free route graph, destination, current access permissions and closures, lift service state, official accessibility assistance route and only the mobility or low-stimulation preferences the student chooses to provide.

Analysis required

Accessible route recommendation: first remove every route that fails the declared access requirement; then estimate current crowding and broad sound conditions on remaining routes; rank by the student's priorities and provide a verified fallback and recheck time.

Answer boundary

Accessibility is a hard constraint, not a weighted preference. Do not infer disability, retain sensitive needs unnecessarily or expose them to others. If any required lift, door or route segment is unverified, do not label the route step-free; provide the official assistance contact.

Why this is useful: Supports a more usable route for taught postgraduates with mobility or sensory preferences.

PGT-062

CORE

L1 - lift status and fallback

R2 - integration-dependent

Taught postgraduate question: Is the lift I need working now, and what step-free alternative should I use if it is not?

Sensors and telemetry

Required operational evidence: authorised lift-controller run/fault state, maintenance outage status and relevant entrance/door/closure state; optional aggregate route footfall. Environmental or movement sensors cannot establish lift availability. Use current source timestamps and service health. Use current official status and instrumented route segments; an unobserved segment remains unknown.

Authoritative supporting data

Authoritative current sources: current accessible-route graph, destination, lift service and maintenance system, temporary closures, building support contact and any authorised assistance procedure.

Analysis required

Authoritative lookup plus fallback routing: report the lift state, source and timestamp; calculate only routes whose every step-free segment is verified; if no route remains, state that clearly and present the official assistance process.

Answer boundary

Do not infer lift operation from door noise, movement or past reliability. A stale controller feed is unknown, not available. Official signage and staff instructions override the chatbot, especially during an incident.

Why this is useful: Prevents a student from relying on an inaccessible route and gives a clear fallback.

7. Accessibility, well-being, navigation and live conditions

Questions PGT-063-PGT-064

PGT-063

CORE

L3 - low-stimulation space recommendation

R2 - integration-dependent

Taught postgraduate question: Which accessible student space is likely to be quieter and less crowded, with steady lighting and reasonable temperature and ventilation indicators?

Sensors and telemetry

Required telemetry (proposed unless verified): correctly located in-room numerical sound, task illuminance/flicker where relevant, air temperature/relative humidity, direct NDIR CO2 and anonymous aggregate occupancy; door state and broad route footfall for access context. Use calibrated/commissioned points, current health, room mapping and missingness; no speech recording.

Authoritative supporting data

Authoritative current sources: present PGT-accessible spaces and permitted use, booking/drop-in status, opening hours, verified step-free routes and lift state, current closures and the student's stated low-noise, low-crowding or lighting preferences.

Analysis required

Current observation + recommendation: filter spaces by verified access and any declared accessibility needs; compare numerical sound, aggregate occupancy, lighting stability, temperature and CO2 using correctly located points; rank a primary and backup and state which criteria are unmeasured.

Answer boundary

This is a low-stimulation space recommendation, not a medical or sensory-needs assessment. Do not infer disability or retain preferences unnecessarily. Sound level does not prove privacy or absence of sudden events; invalid room-level coverage requires a limited or no recommendation.

Why this is useful: Helps a student choose a setting that better supports focused postgraduate work.

PGT-064

SITUATIONAL

L2 - quiet-break recommendation

R2 - integration-dependent

Taught postgraduate question: I have two intensive sessions back to back. Where can I take a ten-minute quiet break without making the next room difficult to reach?

Sensors and telemetry

Required evidence: numerical sound and anonymous aggregate occupancy in verified break areas, current corridor footfall and optional room temperature/CO2; lift and door status for the onward route. Use current, correctly mapped sources with explicit gaps and no speech or identity capture. Require verified room and route mapping, synchronised timestamps, health and gap flags; corridor data is contextual and unobserved route segments remain unknown.

Authoritative supporting data

Authoritative current sources: next-session room/time supplied by the student, current PGT-accessible break areas, permitted use, opening, route and closure status, and any step-free requirement.

Analysis required

Short-gap recommendation: calculate the usable break duration after walking and buffer time; filter to authorised spaces; compare current sound/crowding and route reliability; rank a nearby option and tell the student when to leave.

Answer boundary

This is not a health or rest prescription and cannot guarantee a seat or silence. Do not recommend staff-only or unverified spaces. If route or access status is uncertain, prioritise reaching the next session over environmental preference.

Why this is useful: Builds a practical recovery period into an intensive taught-postgraduate day.

7. Accessibility, well-being, navigation and live conditions

Questions PGT-065-PGT-066

PGT-065

CORE

L3 - move-or-stay recommendation

R2 - integration-dependent

Taught postgraduate question: This room feels warm and stuffy. Are the room-level measurements unusual, and what nearby student space could I use for the rest of my study session?

Sensors and telemetry

Required telemetry (proposed unless verified): correctly located in-room air temperature/relative humidity and direct NDIR CO2; PM2.5 and indicative TVOC only where installed and quality-controlled; anonymous occupancy and serving-zone airflow/fan state; comparable data for authorised alternatives. Use current room/HVAC mapping and explicit missingness.

Authoritative supporting data

Authoritative current sources: present room use and access, booking, duration remaining, verified nearby PGT alternatives, route/closure status, facilities reporting process and any operational bands selected by the institution.

Analysis required

Observation, contextual comparison and recommendation: report current values and rate of change; compare with the room baseline and alternatives over the same window; estimate whether conditions are likely to persist; recommend stay, move or report with reasons and uncertainty.

Answer boundary

Perceived warmth or stuffiness is not a diagnosis. CO2 is only a ventilation indicator, and TVOC is not compound-specific. Without valid in-room evidence or an available alternative, provide the reporting route rather than recommend a move. Withhold causal judgement when periods or coverage are not comparable.

Why this is useful: Gives the student a proportionate, evidence-based option rather than a vague complaint.

PGT-066

SITUATIONAL

L3 - glare forecast and recommendation

R3 - research-grade or restricted

Taught postgraduate question: Is this study area likely to develop distracting glare later this afternoon, and which alternative has a lower predicted risk?

Sensors and telemetry

Required telemetry (proposed unless verified): task-plane and, where possible, eye-level illuminance or luminance proxy at representative positions, solar irradiance, blind state, room orientation and cloud/weather input; anonymous occupancy for candidate availability context. Use commissioned placement, aligned time and coverage metadata.

Authoritative supporting data

Authoritative current sources: present PGT-accessible spaces, current furniture/display orientation, blind operability, bookings, opening hours, route and closure status. The May 2021 layout is not evidence of current desk positions or blinds.

Analysis required

Forecast and recommendation: model afternoon solar position and weather, combine it with measured daylight/blind histories, estimate glare-risk windows at representative positions and rank authorised alternatives with prediction intervals and a recheck time.

Answer boundary

Illuminance alone cannot determine all glare or individual visual comfort, and a room-level forecast cannot guarantee a particular seat. Without eye-level/orientation or blind information, report daylight trend only and mark glare comparison not assessable.

Why this is useful: Helps a student avoid an interruption during screen-intensive reading or analysis.

7. Accessibility, well-being, navigation and live conditions

Questions PGT-067-PGT-068

PGT-067

SITUATIONAL

L3 - class-change route forecast

R2 - integration-dependent

Taught postgraduate question: Which open route is likely to avoid the busiest stairs during the next class change?

Sensors and telemetry

Required evidence: privacy-preserving aggregate counts on relevant stairs, corridors and doors; current lift/door/closure status for alternative routes. Verify segment mapping, timestamp, counter health and coverage; do not use identifiable device tracking. Use current official status and instrumented route segments; an unobserved segment remains unknown.

Authoritative supporting data

Authoritative current sources: current authorised route graph, destination and arrival deadline, access permissions, class-change timetable, temporary closures and any step-free requirement supplied by the student.

Analysis required

Near-term forecast and route recommendation: forecast segment demand over the next class change from current counts and scheduled releases; remove inaccessible/restricted routes; compare travel-time distributions and recommend a route with an alternative and refresh point.

Answer boundary

Counts at multi-entry junctions may be uncertain, and an uninstrumented stair is unknown rather than quiet. Do not compromise a step-free requirement or route through restricted areas to reduce crowding.

Why this is useful: Helps a student move reliably between tightly spaced postgraduate sessions.

PGT-068

CORE

L3 - temporary-accessibility recommendation

R2 - integration-dependent

Taught postgraduate question: I have a temporary mobility limitation. Which suitable room and step-free route would make our group meeting least likely to be disrupted?

Sensors and telemetry

Required evidence: authorised lift, entrance and accessible-door status; privacy-preserving route footfall; anonymous room occupancy and room-condition telemetry only for verified candidate rooms. Use current topology, source timestamps and coverage; no personal tracking. Use current official status and instrumented route segments; an unobserved segment remains unknown.

Authoritative supporting data

Authoritative current sources: the temporary access requirement the student chooses to state, current step-free route graph, room access/bookability, capacity, group requirements, closures, assistance contact and an authorised backup room.

Analysis required

Accessible room and route recommendation: treat every stated access requirement as a hard filter; verify route and room; estimate disruption risk from current service state and recent reliability; then rank valid rooms and supply a contingency.

Answer boundary

Do not infer the nature or duration of the limitation or share it without consent. An empty room is not necessarily bookable, and any route with an unverified lift or door cannot be called accessible. If no valid option exists, give the official assistance route.

Why this is useful: Makes group-work planning more inclusive during a temporary change in mobility.

7. Accessibility, well-being, navigation and live conditions

Questions PGT-069-PGT-070

PGT-069

CORE

L4 - answerability explanation

R1 - practical with core feeds

Taught postgraduate question: Why can you not currently tell me whether this room is comfortable or has good air quality?

Sensors and telemetry

Required evidence is the observability metadata itself: installed/proposed sensor type, exact room or served-zone mapping, measurement method, unit, calibration/commissioning, latest timestamp, uptime, missingness and spatial coverage; relevant non-sensor context such as occupancy and HVAC may also be required. Use time-stamped, health-checked sources with explicit provenance and missing-data flags.

Authoritative supporting data

Authoritative current sources: present room use, access and geometry, serving-HVAC relationship, sensor inventory and maintenance, data-quality records, selected comfort or air-quality definition and any user preference voluntarily supplied. The historic plan cannot establish current instrumentation or use.

Analysis required

Answerability explanation: map 'comfortable' and 'good air quality' to the environmental and personal factors required; list which variables, room coverage, calibration, time window and authoritative context are available; distinguish what can be observed from what remains not assessable.

Answer boundary

Do not convert a partial temperature/CO2 reading into a comfort, health or compliance judgement. Personal thermal comfort also depends on radiant temperature, air speed, activity, clothing and individual preference; air quality may require PM, VOC or other context. Missing evidence must remain explicit.

Why this is useful: Builds trust by making the limits of the talking building visible rather than hiding them behind fluent language.

PGT-070

CORE BOUNDARY CASE

L1 - authoritative emergency response

R3 - research-grade or restricted

Taught postgraduate question: There is an alarm or possible emergency. What official instruction should I follow, and where can I find authoritative route information?

Sensors and telemetry

Use only authorised certified alarm/incident-system status and official lift, door or route closures released for occupants. Experimental environmental, acoustic or occupancy sensing is not the primary emergency source and must not be used to infer an all-clear. Use current official status and instrumented route segments; an unobserved segment remains unknown.

Authoritative supporting data

Authoritative current sources: official alarm state and message, evacuation or shelter instruction, building signage/status, University security or emergency contact and any incident-specific authorised route information.

Analysis required

Authoritative emergency lookup: display the latest official instruction, source and timestamp; direct the student to follow alarms, signage, staff and emergency services; provide a route only when the authorised emergency system supplies one.

Answer boundary

Do not diagnose the incident, delay evacuation, override an alarm, declare an area safe or improvise a route from ordinary sensor data. If the official feed is unavailable, instruct the student to follow on-site alarms and staff and contact emergency services as appropriate.

Why this is useful: Provides a clear boundary for the highest-consequence interaction a student may attempt.

8. Sustainability, faults, evidence and responsible use

Questions PGT-071-PGT-072

PGT-071

SITUATIONAL

L3 - low-impact room recommendation

R3 - research-grade or restricted

Taught postgraduate question: For our four-person project meeting, which room is the right size and likely to avoid unnecessary energy use without giving us poorer measured conditions?

Sensors and telemetry

Design requirement: anonymous occupancy; in-room temperature/RH and direct NDIR CO2; lighting and HVAC state; room-, zone- or system-level electricity/heat metering with current topology. Use aligned, quality-flagged intervals, exclude unresolved gaps and make shared-load allocation explicit.

Authoritative supporting data

Authoritative current sources: current four-person-appropriate rooms, access and booking, capacity and permitted use, meeting duration, accessibility, required services, HVAC/meter relationships and operating rules. The May 2021 plan is not evidence of current room or meter topology.

Analysis required

Multi-criteria recommendation + bounded energy calculation: apply access, booking, capacity, activity and accessibility first; estimate marginal or allocated energy only where meter/HVAC topology supports it; compare measured-condition risk; rank options and show shared loads and uncertainty.

Answer boundary

A smaller, empty-looking or warmer room is not automatically lower energy. Shared HVAC and baseloads may prevent room attribution. Never trade away access, safety or required service for energy savings; if metering is inadequate, omit the energy criterion.

Why this is useful: Lets students make a practical sustainability choice without sacrificing a usable working environment.

PGT-072

ADVANCED

L4 - after-hours waste analysis

R3 - research-grade or restricted

Taught postgraduate question: During deadline week, which empty areas appear to be lit, heated, cooled or ventilated when they may not need to be?

Sensors and telemetry

Required (proposed unless verified): anonymous aggregate occupancy; lighting state and illuminance; HVAC fan/airflow, valve/damper and zone call; room/zone temperature; electricity or heat metering with verified serving topology. Use synchronised, quality-flagged streams and documented operating schedules; do not identify occupants.

Authoritative supporting data

Authoritative current sources: student-project data permission, verified monitored spaces, current room use, booking/timetable, opening and cleaning/maintenance schedules, HVAC/lighting control intent, meter topology and exclusions for safety, fabric protection or required ventilation.

Analysis required

Historical exception analysis: identify periods with high-confidence vacancy; compare observed lighting/HVAC operation with authorised schedules and control requirements; calculate duration and, only where metering supports it, associated energy; rank candidate cases for investigation.

Answer boundary

Vacancy estimates and control state do not prove waste. Exclude mandatory ventilation, frost protection, cleaning, maintenance and sensor uncertainty. Do not expose room-use patterns at a level that creates privacy or security risk; report 'not attributable' when meter topology is inadequate.

Why this is useful: Gives a taught postgraduate a defensible sustainability-project question that links occupancy, controls and energy without labelling every after-hours operation as waste.

8. Sustainability, faults, evidence and responsible use

Questions PGT-073-PGT-074

PGT-073

SITUATIONAL

L4 - intervention evaluation

R3 - research-grade or restricted

Taught postgraduate question: After the ventilation controls were changed, did energy use fall without worsening CO2 or temperature during similar classes?

Sensors and telemetry

Required telemetry: room temperature and NDIR CO2, anonymous occupancy, serving-zone airflow/fan/damper/valve state, and electricity/heat metering with verified topology; outdoor weather. Preserve intervention date, control version, calibration/commissioning, gaps and topology changes. Require verified room and meter topology, aligned timestamps, health and gap flags; corridor data is contextual and shared loads must be explicit.

Authoritative supporting data

Authoritative current sources: approved project access, intervention specification and date, affected rooms/equipment, comparable class timetable and expected attendance, maintenance/override records, operating targets and any control spaces.

Analysis required

Intervention evaluation: select comparable pre/post sessions and, where possible, unaffected control rooms; normalise for occupancy, weather and schedule; estimate changes in energy, CO2 and temperature with uncertainty; test whether gains persist and report trade-offs.

Answer boundary

A simple before/after average cannot establish the intervention effect. Do not claim causality when controls, occupancy, weather, metering or room use changed simultaneously. Apply data permissions and return a limited result when no credible comparison exists.

Why this is useful: Supports a rigorous taught-postgraduate building-performance assignment rather than a superficial dashboard comparison.

PGT-074

SITUATIONAL

L4 - session-energy attribution

R3 - research-grade or restricted

Taught postgraduate question: Roughly how much building energy can reasonably be associated with our late-evening study session compared with a similar daytime session?

Sensors and telemetry

Required telemetry: room/zone electricity and, where relevant, thermal-energy meters; HVAC, lighting and plug-load state; anonymous occupancy; outdoor weather. Use verified meter boundaries, allocation topology, clock alignment and missing-data flags. Require current meter topology, aligned intervals and explicit treatment of shared loads and gaps.

Authoritative supporting data

Authoritative current sources: project permission, session room/time/duration, current room use, operating schedules, comparable daytime sessions, tariff/carbon factors only if required, and exclusions for shared or mandatory loads.

Analysis required

Bounded calculation: construct a counterfactual from comparable occupied and unoccupied periods; separate measured, allocated and modelled loads; estimate an energy range attributable to the session and show shared baseload/HVAC uncertainty.

Answer boundary

Do not present whole-floor or shared HVAC consumption as the group's exact energy use. Avoid named attendance data and moral judgements. If submetering or allocation topology is inadequate, report the directly measured zone/building change and mark session attribution not assessable.

Why this is useful: Teaches students how to make a transparent energy-attribution estimate without overstating what building meters can resolve.

8. Sustainability, faults, evidence and responsible use

Questions PGT-075-PGT-076

PGT-075

CORE

L1 - issue reporting and status

R2 - integration-dependent

Taught postgraduate question: I have found a leak, overflowing waste point or damaged fitting. What is the official reporting route, and can I see a privacy-safe acknowledgement status?

Sensors and telemetry

Required evidence may include a time-stamped point/cable leak detector, water-flow anomaly, bin-fill or local equipment fault where deployed, but the student report and authorised maintenance workflow remain primary. Use current sensor health, exact location mapping and event provenance; no camera or identity is required by default.

Authoritative supporting data

Authoritative current sources: current room/location naming and access, official facilities or waste reporting channel, incident category, priority and safety instructions, work-order acknowledgement state and permission rules for what status may be shown to the reporter.

Analysis required

Authoritative issue reporting + status lookup: classify the reported location and issue, attach only necessary evidence, submit through the official channel, return a case reference, and expose only permitted acknowledgement states such as received, assigned or closed.

Answer boundary

Do not publish the reporter's identity, other users' reports, security-sensitive detail or maintenance notes. For an immediate hazard, leak near electricity or emergency condition, direct the student to on-site staff/security/emergency procedures rather than waiting for the catalogue service.

Why this is useful: Makes it easier for students to report sustainability and maintenance problems and know that they entered the correct process.

PGT-076

CORE

L4 - conflicting-evidence diagnosis

R2 - integration-dependent

Taught postgraduate question: The room display and the talking-building answer show different temperature or CO2 values. Which one should I rely on, and how will the conflict be resolved?

Sensors and telemetry

Required evidence: both displayed and queried source identifiers, exact observed property, unit, room/zone mapping, sampling and display aggregation, latest raw timestamp, calibration/commissioning, health, missingness and any transformation or rounding. Preserve independent provenance for each value. Use time-stamped, health-checked sources with explicit provenance and missing-data flags.

Authoritative supporting data

Authoritative current sources: current sensor/display configuration, unit and timezone, data pipeline and cache status, maintenance or replacement history, room mapping and the official source-of-truth policy for that service.

Analysis required

Conflict diagnosis: compare source identity, sensor location, observed property, unit, timestamp, averaging window, calibration/commissioning and transformation logic; determine whether the values are genuinely inconsistent, differently aggregated or stale; select the authoritative display source where policy defines one.

Answer boundary

Do not choose the numerically convenient value or hide disagreement. A corridor sensor and in-room point are not interchangeable. If source metadata or calibration is insufficient, report the conflict unresolved and direct it to the data owner.

Why this is useful: Helps students understand why building data can disagree and prevents false precision.

8. Sustainability, faults, evidence and responsible use

Questions PGT-077-PGT-078

PGT-077

CORE

L4 - recommendation confidence query

R1 - practical with core feeds

Taught postgraduate question: How confident is this room recommendation, which evidence is weakest, and when should I recheck it?

Sensors and telemetry

Required evidence is the complete provenance bundle for the recommendation: source IDs, room/route mapping, units, timestamps, calibration or service health, uptime, missingness, spatial coverage and forecast model/version for every criterion used. Use time-stamped, health-checked sources with explicit provenance and missing-data flags.

Authoritative supporting data

Authoritative current sources: candidate access/bookability, current service states, requested constraints, comparison history, model validation results, permission rules and the time at which the recommendation expires or must be refreshed.

Analysis required

Confidence explanation: decompose the recommendation into hard constraints, observations, lookups, calculations and forecasts; report coverage, freshness, missingness, model error, source conflict, sensitivity to preferences and the condition that should trigger a recheck or switch to the backup.

Answer boundary

A confidence label is not a probability of personal satisfaction unless it is calibrated for that outcome. Do not hide missing criteria inside a single score. Withdraw any component that lacks permitted provenance or valid coverage.

Why this is useful: Lets a postgraduate decide whether to rely on the answer for a high-value study or assessed activity.

PGT-078

SITUATIONAL

L4 - repair verification

R2 - integration-dependent

Taught postgraduate question: A recurring temperature or ventilation problem was marked as fixed. Do comparable occupied periods show that conditions have returned to their normal range?

Sensors and telemetry

Required telemetry: the same correctly located room and serving-system variables that established the fault, plus equipment command/feedback and relevant energy where appropriate; stable pre/post sensor placement, calibration, clock alignment, health and coverage are essential. Require current meter topology, aligned intervals and explicit treatment of shared loads and gaps.

Authoritative supporting data

Authoritative current sources: exact work-order scope and completion time, asset/room relationships, operating state, timetable/occupancy, weather, setpoints, overrides and later recurrence reports. Current topology must be verified operationally.

Analysis required

Repair verification: compare time-matched occupied periods before and after the recorded intervention, adjusting for occupancy, weather, schedule and operating state; test persistence, not one favourable reading; report effect size, uncertainty and remaining anomalies.

Answer boundary

The work-order status alone does not prove restoration, and improved room conditions do not prove which repair caused them. If pre/post periods or sensor configuration are not comparable, report the evidence as inconclusive and keep the official maintenance route.

Why this is useful: Gives students transparent evidence about whether a repeatedly reported problem has improved.

8. Sustainability, faults, evidence and responsible use

Questions PGT-079-PGT-080

PGT-079

CORE

L2 - provenance and calibration query

R1 - practical with core feeds

Taught postgraduate question: What sensors, building systems and official data sources support this answer, where are they located, and how recent and reliable are they?

Sensors and telemetry

Required evidence is provenance rather than a new measurement: sensor or system identifier, observed property, unit, exact room/served zone, mounting or endpoint, sampling/aggregation, latest timestamp, calibration/commissioning, health, missingness and transformation history for every source used. Use aggregate service or endpoint health only; do not inspect content, credentials or identifiable traffic.

Authoritative supporting data

Authoritative current sources: sensor and asset registry, ontology/knowledge-graph relationships, BMS or service inventory, data catalogue, calibration and maintenance records, access-control policy and the version of any threshold, model or external document applied.

Analysis required

Provenance lookup: list every permitted sensor/system source, observed property, room or served zone, timestamp, sampling/aggregation window, calibration or commissioning state, current health, missingness and transformation used in the answer; identify indirect or omitted evidence.

Answer boundary

Do not reveal credentials, connection strings, exact security-system topology, personal data or unauthorised maintenance detail. If a claim has no permitted traceable source, remove or qualify it; provenance completeness is part of answerability.

Why this is useful: Allows students to inspect the basis of an analytical answer rather than trusting fluent text alone.

PGT-080

CORE BOUNDARY CASE

L4 - high-consequence clarification

R3 - research-grade or restricted

Taught postgraduate question: Can you tell me whether this room is safe or compliant to use, or do I need to specify the particular hazard or requirement?

Sensors and telemetry

No generic sensor package can answer this question. Evidence must be selected only after the student names the hazard or requirement: certified alarm or specialist detector status, calibrated environmental measurement, authoritative access/closure state or another competent-system source. Preserve source authority, location, timestamp, quality and coverage.

Authoritative supporting data

Authoritative current sources depend on the clarified issue: applicable Welsh or University requirement, current room use, risk assessment, certified safety system, inspection/maintenance record, official incident message, competent owner and student permission. The May 2021 plan is not compliance evidence.

Analysis required

Clarification + evidence-obligation assessment: ask whether the concern is fire, gas, electrical, air quality, access, occupancy, specialist activity or a named requirement; map the clarified claim to the correct authoritative systems, measurements, spatial/temporal coverage and competent owner; return observation/lookup, supported assessment, conflicting evidence or not assessable.

Answer boundary

Never answer broad safety or compliance from general environmental IoT data or an LLM judgement. Do not delay evacuation or override official instructions. For an immediate hazard, follow alarms, staff and emergency services; when authority or evidence is incomplete, return not assessable and provide the competent contact.

Why this is useful: Prevents a high-consequence student question from receiving a generic answer and directs it to the correct evidence and authority.

4. Catalogue summary and implementation requirements

The catalogue combines practical taught-postgraduate service questions with integration-dependent forecasts, multi-source assessments and restricted safety cases. Readiness identifies the verified evidence and controls required for a responsible answer.

Questions	80	Eight taught-postgraduate-centred sections containing ten questions each.
R1 practical	6	Can be supported with proposed core sensing and basic authoritative spatial or service feeds.
R2 integration-dependent	52	Require timetable, booking, access, IT, AV, lift, BMS, maintenance or forecasting integration.
R3 research or restricted	22	Require dense sensing, validated diagnosis, controlled operational data, or stronger privacy and safety governance.

Recurring system capabilities

- 1

Time-aware academic planning
Resolve intensive timetables, short gaps, deadline peaks, commuting windows and transitions between different kinds of work.
- 2

Room and route reasoning
Use current rooms, access, bookings, capacities, permitted activities, step-free routes, lift status and closures.
- 3

Environmental and service evidence
Combine correctly located room sensors with anonymous occupancy, BMS, energy, network, AV and specialist-service status.
- 4

Forecast, comparison and diagnosis
Predict demand and room conditions, build matched baselines, rank alternatives and separate association from supported diagnosis.
- 5

Evidence quality and provenance
Check calibration, timestamps, missingness, spatial coverage, source conflicts, permissions and whether each claim is direct or inferred.
- 6

Clarification, privacy and abstention
Ask for missing constraints, minimise schedule and personal data, refuse unauthorised requests and return not assessable when evidence is inadequate.

QUESTION RECORD STRUCTURE

Each entry can become a system test containing the natural utterance and paraphrases, required entities, minimum room and service evidence, expected operation type, authorisation rule, answerability class, acceptable clarification, uncertainty statement, expected answer characteristics and prohibited overclaims.

5. Evidence, permissions and answerability rules

These rules apply across all 80 records and define when the building may answer, qualify, ask for clarification or return not assessable.

Building and room status

The May 2021 plan is a historical spatial reference. Current room purpose, capacity, taught-postgraduate access, bookability, equipment, entrances, amenities, lifts, services and closures require an authoritative current source.

Spatial coverage

A room-level answer requires a correctly located sensor inside that room or a validated served zone. Corridor and adjacent-room data may be labelled context but cannot be silently substituted.

Time window and operation type

Every answer resolves the requested period and labels its operation as observation, lookup, calculation, estimate, forecast, diagnosis or recommendation. Stale evidence is not current status.

Calibration and data quality

Use commissioning or calibration records, uptime, missingness, drift, source agreement and coverage. A failed or unverified sensor cannot support a precise comparison, forecast, diagnosis or compliance claim.

Occupancy, workstations and privacy

Use anonymous aggregate counts. Occupancy does not identify people or guarantee a seat, room, socket or workstation; bookings, access and endpoint service status remain authoritative.

Sound, meetings and speech

Routine acoustic services use numerical LAeq and percentile statistics. Do not retain speech, infer conversation content or claim privacy, intelligibility or disturbance to an individual from sound level alone.

Environmental interpretation

Direct-reading NDIR CO2 is a ventilation indicator, not complete air quality. Temperature, RH, PM, indicative TVOC, airflow, radiant conditions, outdoor weather and activity may be relevant subject to coverage and quality.

Specialist rooms and safety

Environmental IoT data cannot authorise a laboratory activity or certify safety. Current room function, induction, supervision, risk assessment, equipment and official procedures come from authoritative institutional systems.

Operational services and permissions

Timetables, bookings, room use, access, lifts, network, AV, printing, maintenance and closures come from authorised feeds. Enforce the student role and minimise personal and schedule data.

Emergency and high-consequence answers

Certified alarms and official communications take precedence. Broad safety, health and compliance questions require clarification and competent authority; missing or conflicting critical evidence requires a limited or not-assessable answer.

FINAL ANSWERABILITY RULE

Answer only when the required variables, current spatial coverage, requested time window, source authority, data quality and permissions are adequate. Otherwise ask for clarification, omit unsupported criteria, provide a limited answer with explicit uncertainty, or return not assessable.